THE NEGATIVE-VISCOSITY EFFECT IN FERROFLUIDS

A Complete First-Principles Reconstruction of Every Equation

VOLUME II

Statistical Mechanics of the Orientations,
Rotational Brownian Motion,
and Equations (SM-2)–(SM-5)

Where Volume I ended. Volume I built the mathematical apparatus (index notation, vector calculus, the local decomposition $\boldsymbol{\Omega} = \frac{1}{2}\nabla \times \boldsymbol{v}$), the continuum equations (continuity, Cauchy, Navier–Stokes), and the Stokes-flow result $\zeta_r = 8\pi\eta a^3 = 6\eta V = \pi\eta d^3$ for the rotational friction of a sphere. It closed with the eight-part derivation of

$$\tau_s = \frac{\rho_s d^2}{60\eta} = 10^{-11}\ \text{s}, \qquad \text{and the identity} \qquad \frac{I}{\tau_s} = n\zeta_r = 6\eta\varphi. \tag{SM-1}$$

Volume II now supplies the three missing ingredients announced in Sec. 7.9 — the Boltzmann/Langevin statistics of the moments, rotational Brownian motion, and the link between magnetic torque and viscosity — and then derives Eqs. (SM-2), (SM-3), (SM-4) and (SM-5) in the papers' own order. All numbering, notation and conventions of Volume I continue unchanged; chapter numbers resume at 9, page numbers at 34.

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Part IV

Statistical Mechanics of the Orientations

Chapter 9

The Boltzmann Distribution

9.1 Why a distribution is needed at all

A magnetic grain of diameter $d = 10$ nm carries a magnetic moment $m = M_s V$ which, in a field H, has the orientational energy $-mH\cos\theta$ of Eq. (4.4.2). If nothing else acted, every moment would simply align with the field and the suspension would be saturated at any field strength whatsoever. That is not what happens: the magnetisation of a ferrofluid rises gradually with H over a range of hundreds of oersteds. The reason is that the grain is immersed in a liquid whose molecules bombard it $\sim 10^{12}$ times per second from random directions, delivering random torques that continually knock the moment away from the field direction. The competition is between two energies:

$$\underbrace{mH}_{\text{magnetic}} \qquad \text{versus} \qquad \underbrace{kT}_{\text{thermal}}, \tag{9.1.1}$$

whose ratio is exactly the Langevin parameter $\xi = mH/kT$ defined in Eq. (4.4.6). We therefore need the statistical law that says how often a given orientation occurs. That law is the Boltzmann distribution, which we now derive.

9.2 The two postulates of statistical mechanics

Assumption 9.2.1 (Equal a priori probabilities)**.** An isolated system in equilibrium is equally likely to be found in each of the microstates accessible to it at its given energy.

Assumption 9.2.2 (Boltzmann's entropy)**.** The entropy of an isolated system with $\mathcal{N}(E)$ accessible microstates at energy E is

$$S(E) = k\ln\mathcal{N}(E). \tag{9.2.1}$$

These two statements cannot be derived from mechanics; they are the postulates on which statistical mechanics rests, and we adopt them. Everything below *is* derived.

We also need the thermodynamic definition of absolute temperature. Consider two systems, with energies E_1 and E_2 and $E_1 + E_2 = E_{\text{tot}}$ fixed, allowed to exchange energy. The number of joint microstates is $\mathcal{N}_1(E_1)\mathcal{N}_2(E_{\text{tot}} - E_1)$, and by Assumption 9.2.1 the observed value of E_1 is overwhelmingly the one that maximises this product, i.e. that maximises $S_1(E_1)+S_2(E_{\text{tot}}-E_1)$. Setting the derivative to zero,

$$\frac{\partial S_1}{\partial E_1} - \frac{\partial S_2}{\partial E_2} = 0 \qquad \Longrightarrow \qquad \frac{\partial S_1}{\partial E_1} = \frac{\partial S_2}{\partial E_2}. \tag{9.2.2}$$

So two systems in thermal equilibrium share the common value of $\partial S/\partial E$. This is precisely the property a thermometer measures, and we *define*

$$\frac{1}{T} \equiv \frac{\partial S}{\partial E}. \tag{9.2.3}$$

9.3 Derivation of the Boltzmann factor

Let a small system $\mathcal{A}$ (here: one magnetic grain, whose only relevant variable is the direction $\boldsymbol{e}$ of its moment) be in contact with a large reservoir $\mathcal{R}$ (the carrier liquid) at temperature T. The compound system is isolated, with total energy E_{tot}.

Step 1. Suppose $\mathcal{A}$ is in one definite microstate i of energy E_i. Then the reservoir has energy $E_{\mathrm{tot}} - E_i$ and $\mathcal{N}_{\mathcal{R}}(E_{\mathrm{tot}} - E_i)$ microstates available. By Assumption 9.2.1 applied to the *compound* system, the probability of finding $\mathcal{A}$ in state i is proportional to the number of ways the rest of the world can accommodate it:

$$p_i \propto \mathcal{N}_{\mathcal{R}}(E_{\mathrm{tot}} - E_i). \tag{9.3.1}$$

Step 2. Take the logarithm and use (9.2.1):

$$\ln p_i = \mathrm{const} + \frac{1}{k} S_{\mathcal{R}}(E_{\mathrm{tot}} - E_i). \tag{9.3.2}$$

Step 3. Expand in E_i. This is legitimate because the reservoir is enormous: $E_i \ll E_{\mathrm{tot}}$, so E_i is a tiny displacement of the reservoir's energy and a first-order Taylor expansion is accurate:

$$S_{\mathcal{R}}(E_{\mathrm{tot}} - E_i) = S_{\mathcal{R}}(E_{\mathrm{tot}}) - E_i \left.\frac{\partial S_{\mathcal{R}}}{\partial E}\right|_{E_{\mathrm{tot}}} + O(E_i^2) \tag{9.3.3}$$

$$= S_{\mathcal{R}}(E_{\mathrm{tot}}) - \frac{E_i}{T} + O(E_i^2), \tag{9.3.4}$$

where the definition (9.2.3) was used. The neglected term involves $\partial^2 S/\partial E^2 = -(1/T^2)\partial T/\partial E$, which vanishes for a reservoir so large that absorbing E_i does not change its temperature. That is the defining property of a reservoir.

Step 4. Therefore

$$\ln p_i = \mathrm{const}' - \frac{E_i}{kT} \qquad \Longrightarrow \qquad \boxed{p_i = \frac{1}{Z}\, e^{-E_i/kT}} \tag{9.3.5}$$

with Z (the partition function) fixed by $\sum_i p_i = 1$. ∎

Note what (9.3.5) says physically: states of *lower* energy are exponentially more probable, and the scale on which "lower" is measured is kT. Energy differences much smaller than kT are essentially invisible; differences much larger than kT are decisive.

9.4 Orientations: the solid-angle measure

The magnetic moment of a rigid dipole is specified by a unit vector $\boldsymbol{e}$, i.e. by a point on the unit sphere. In spherical polar coordinates with the polar axis along the applied field $\hat{\boldsymbol{h}}$,

$$\boldsymbol{e} = (\sin\theta\cos\phi,\ \sin\theta\sin\phi,\ \cos\theta), \qquad 0 \le \theta \le \pi, \quad 0 \le \phi < 2\pi, \tag{9.4.1}$$

where θ is the angle between $\boldsymbol{e}$ and $\hat{\boldsymbol{h}}$, so that $\boldsymbol{e}\cdot\hat{\boldsymbol{h}} = \cos\theta$.

The set of orientations is continuous, so instead of probabilities p_i we need a probability *density* $W(\boldsymbol{e})$ such that $W\,\mathrm{d}\Omega_e$ is the probability of finding the moment within the element of solid angle $\mathrm{d}\Omega_e$. We must therefore know $\mathrm{d}\Omega_e$.

Derivation of $\mathrm{d}\Omega_e = \sin\theta\,\mathrm{d}\theta\,\mathrm{d}\phi$. On a sphere of radius 1, increasing θ by $\mathrm{d}\theta$ moves a point a distance $\mathrm{d}\theta$ along a meridian (arc length = radius × angle = $1\cdot\mathrm{d}\theta$). Increasing ϕ by $\mathrm{d}\phi$ moves it along a circle of latitude whose radius is not 1 but $\sin\theta$ (the perpendicular distance from the polar axis), so the arc length is $\sin\theta\,\mathrm{d}\phi$. The two displacements are perpendicular, so the patch is a rectangle of area

$$\mathrm{d}\Omega_e = \mathrm{d}\theta\cdot\sin\theta\,\mathrm{d}\phi = \sin\theta\,\mathrm{d}\theta\,\mathrm{d}\phi. \tag{9.4.2}$$

Check: $\int \mathrm{d}\Omega_e = \int_0^{2\pi}\mathrm{d}\phi\int_0^{\pi}\sin\theta\,\mathrm{d}\theta = 2\pi\cdot\big[-\cos\theta\big]_0^{\pi} = 2\pi\cdot 2 = 4\pi$, the total solid angle. ✓

It is almost always convenient to substitute

$$u \equiv \cos\theta, \qquad \mathrm{d}u = -\sin\theta\,\mathrm{d}\theta, \tag{9.4.3}$$

which converts $\int_0^{\pi}(\cdots)\sin\theta\,\mathrm{d}\theta$ into $\int_{-1}^{1}(\cdots)\,\mathrm{d}u$ — the sign from $\mathrm{d}u$ being cancelled by the reversal of the limits ($\theta=0\Rightarrow u=1$, $\theta=\pi\Rightarrow u=-1$).

9.5 The equilibrium orientational distribution

Insert the dipole energy $U = -\boldsymbol{m}\cdot\boldsymbol{H} = -mH\cos\theta$ of Eq. (4.4.2) into the Boltzmann law (9.3.5):

$$W_0(\boldsymbol{e}) = C\exp\!\left(\frac{mH\cos\theta}{kT}\right) = C\,e^{\xi\cos\theta} = C\,e^{\xi\,\boldsymbol{e}\cdot\hat{\boldsymbol{h}}}, \tag{9.5.1}$$

with $\xi = mH/kT$ as defined in (4.4.6). The subscript 0 marks the *equilibrium* distribution, i.e. the one holding when the fluid is at rest; all of Volume II's later work consists of computing the small correction to (9.5.1) produced by flow.

Normalisation. Require $\int W_0\,\mathrm{d}\Omega_e = 1$:

$$1 = C\int_0^{2\pi}\mathrm{d}\phi\int_0^{\pi} e^{\xi\cos\theta}\sin\theta\,\mathrm{d}\theta \tag{9.5.2}$$

$$= 2\pi C\int_{-1}^{1} e^{\xi u}\,\mathrm{d}u \qquad \text{by (9.4.3)} \tag{9.5.3}$$

$$= 2\pi C\left[\frac{e^{\xi u}}{\xi}\right]_{-1}^{1} \tag{9.5.4}$$

$$= 2\pi C\,\frac{e^{\xi}-e^{-\xi}}{\xi} \tag{9.5.5}$$

$$= \frac{4\pi C\sinh\xi}{\xi}, \qquad \text{using } \sinh\xi = \tfrac{1}{2}(e^{\xi}-e^{-\xi}) \tag{9.5.6}$$

so that

$$C = \frac{\xi}{4\pi\sinh\xi}, \qquad W_0(\theta) = \frac{\xi}{4\pi\sinh\xi}\,e^{\xi\cos\theta}. \tag{9.5.7}$$

Check the field-free limit. As $\xi\to 0$, $\sinh\xi\to\xi$ and $e^{\xi\cos\theta}\to 1$, so $W_0\to 1/4\pi$: all directions equally likely, as they must be when there is no field. ✓

The integral appearing above will recur so often that we name it:

$$Z(\xi) \equiv \int e^{\xi\, \boldsymbol{e}\cdot\hat{\boldsymbol{h}}}\, \mathrm{d}\Omega_e = \frac{4\pi \sinh \xi}{\xi}. \tag{9.5.8}$$

Chapter 10

The Langevin Function and the Equilibrium Magnetisation

10.1 Context

The magnetisation of the suspension was defined in Eq. (4.5.1) as $\boldsymbol{M} = nm\langle \boldsymbol{e} \rangle$. With the distribution (9.5.7) in hand we can now compute $\langle \boldsymbol{e} \rangle$ exactly. The result is the function called $L(\xi)$ in both papers, and it enters Eq. (SM-10), Eq. (B-4) and, through them, essentially every later formula.

10.2 Computation of $\langle \boldsymbol{e} \rangle$

Step 1: only the component along the field survives. By symmetry the distribution (9.5.7) depends on θ but not on ϕ. The components $e_x = \sin\theta\cos\phi$ and $e_y = \sin\theta\sin\phi$ therefore average to zero, since $\int_0^{2\pi} \cos\phi \, \mathrm{d}\phi = \int_0^{2\pi} \sin\phi \, \mathrm{d}\phi = 0$. Hence $\langle \boldsymbol{e} \rangle = \langle \cos\theta \rangle \, \hat{\boldsymbol{h}}$, and only one integral is needed.

Step 2: set up the integral.

$$\langle \cos\theta \rangle = \int \cos\theta \, W_0 \, \mathrm{d}\Omega_e = \frac{\displaystyle\int_0^{2\pi} \mathrm{d}\phi \int_0^{\pi} \cos\theta \, e^{\xi\cos\theta} \sin\theta \, \mathrm{d}\theta}{\displaystyle\int_0^{2\pi} \mathrm{d}\phi \int_0^{\pi} e^{\xi\cos\theta} \sin\theta \, \mathrm{d}\theta} \tag{10.2.1}$$

$$= \frac{\displaystyle\int_{-1}^{1} u \, e^{\xi u} \, \mathrm{d}u}{\displaystyle\int_{-1}^{1} e^{\xi u} \, \mathrm{d}u}. \tag{10.2.2}$$

The factors 2π and C cancel between numerator and denominator, which is why writing the average as a ratio of unnormalised integrals is safe.

Step 3: the denominator. Already computed:

$$\mathcal{D} \equiv \int_{-1}^{1} e^{\xi u} \, \mathrm{d}u = \frac{e^{\xi} - e^{-\xi}}{\xi} = \frac{2\sinh\xi}{\xi}. \tag{10.2.3}$$

Step 4: the numerator, by the differentiation trick. Notice that

$$u\,e^{\xi u} = \frac{\partial}{\partial \xi} e^{\xi u}, \tag{10.2.4}$$

because differentiating $e^{\xi u}$ with respect to the *parameter* ξ brings down a factor u. Since the limits of integration do not depend on ξ, differentiation and integration may be interchanged, and

$$\mathcal{N} \equiv \int_{-1}^{1} u\,e^{\xi u}\,\mathrm{d}u = \frac{\mathrm{d}}{\mathrm{d}\xi}\int_{-1}^{1} e^{\xi u}\,\mathrm{d}u = \frac{\mathrm{d}}{\mathrm{d}\xi}\left(\frac{2\sinh\xi}{\xi}\right) \tag{10.2.5}$$

$$= 2\left[\frac{\cosh\xi}{\xi} - \frac{\sinh\xi}{\xi^2}\right] \qquad \text{(quotient rule; } \sinh' = \cosh\text{)} \tag{10.2.6}$$

$$= \frac{2(\xi\cosh\xi - \sinh\xi)}{\xi^2}. \tag{10.2.7}$$

Step 4′: the same numerator by integration by parts (independent check). With $\mathrm{d}v = e^{\xi u}\mathrm{d}u$, $v = e^{\xi u}/\xi$, and the other factor u:

$$\int_{-1}^{1} u\,e^{\xi u}\,\mathrm{d}u = \left[\frac{u\,e^{\xi u}}{\xi}\right]_{-1}^{1} - \int_{-1}^{1}\frac{e^{\xi u}}{\xi}\,\mathrm{d}u \tag{10.2.8}$$

$$= \frac{e^{\xi} + e^{-\xi}}{\xi} - \frac{1}{\xi}\cdot\frac{e^{\xi} - e^{-\xi}}{\xi} \tag{10.2.9}$$

$$= \frac{2\cosh\xi}{\xi} - \frac{2\sinh\xi}{\xi^2} = \frac{2(\xi\cosh\xi - \sinh\xi)}{\xi^2}. \quad \checkmark \tag{10.2.10}$$

(In the second line the boundary term was evaluated as $\frac{1}{\xi}[1\cdot e^{\xi} - (-1)e^{-\xi}] = (e^{\xi}+e^{-\xi})/\xi$.)

Step 5: the ratio.

$$\langle\cos\theta\rangle = \frac{\mathcal{N}}{\mathcal{D}} = \frac{2(\xi\cosh\xi - \sinh\xi)/\xi^2}{2\sinh\xi/\xi} \tag{10.2.11}$$

$$= \frac{\xi\cosh\xi - \sinh\xi}{\xi^2}\cdot\frac{\xi}{\sinh\xi} \tag{10.2.12}$$

$$= \frac{\xi\cosh\xi - \sinh\xi}{\xi\,\sinh\xi} \tag{10.2.13}$$

$$= \frac{\cosh\xi}{\sinh\xi} - \frac{1}{\xi} \tag{10.2.14}$$

$$= \coth\xi - \frac{1}{\xi}. \tag{10.2.15}$$

Result 10.2.1 (Langevin function)**.**

$$\boxed{\langle\cos\theta\rangle = L(\xi) \equiv \coth\xi - \frac{1}{\xi}} \tag{10.2.16}$$

10.3 Properties of $L(\xi)$ that will be needed

(a) Weak fields: the series. We need $\coth\xi$ for small ξ. Use the power series of the hyperbolic functions (each obtained from the exponential series):

$$\sinh\xi = \xi + \frac{\xi^3}{6} + \frac{\xi^5}{120} + \cdots, \qquad \cosh\xi = 1 + \frac{\xi^2}{2} + \frac{\xi^4}{24} + \cdots \tag{10.3.1}$$

Then

$$\coth\xi = \frac{\cosh\xi}{\sinh\xi} = \frac{1+\dfrac{\xi^2}{2}+\dfrac{\xi^4}{24}}{\xi\left(1+\dfrac{\xi^2}{6}+\dfrac{\xi^4}{120}\right)} \tag{10.3.2}$$

$$= \frac{1}{\xi}\,(1+a)\,(1+b)^{-1}, \qquad a=\frac{\xi^2}{2}+\frac{\xi^4}{24}, \quad b=\frac{\xi^2}{6}+\frac{\xi^4}{120} \tag{10.3.3}$$

$$= \frac{1}{\xi}\,(1+a)\,\big(1-b+b^2-\cdots\big) \qquad \text{geometric series, } |b|<1 \tag{10.3.4}$$

$$= \frac{1}{\xi}\Big(1+(a-b)+(b^2-ab)+O(\xi^6)\Big). \tag{10.3.5}$$

Now evaluate the brackets term by term:

$$a-b = \left(\frac{\xi^2}{2}-\frac{\xi^2}{6}\right)+\left(\frac{\xi^4}{24}-\frac{\xi^4}{120}\right) = \frac{\xi^2}{3}+\frac{\xi^4}{30}, \tag{10.3.6}$$

$$b^2-ab = \frac{\xi^4}{36}-\frac{\xi^2}{2}\cdot\frac{\xi^2}{6}+O(\xi^6) = \frac{\xi^4}{36}-\frac{\xi^4}{12} = -\frac{\xi^4}{18}, \tag{10.3.7}$$

where $\frac{1}{2}-\frac{1}{6}=\frac{1}{3}$ and $\frac{1}{24}-\frac{1}{120}=\frac{5-1}{120}=\frac{1}{30}$, and $\frac{1}{36}-\frac{1}{12}=\frac{1-3}{36}=-\frac{1}{18}$. Adding the two ξ^4 pieces, $\frac{1}{30}-\frac{1}{18}=\frac{3-5}{90}=-\frac{1}{45}$, so

$$\coth\xi = \frac{1}{\xi}+\frac{\xi}{3}-\frac{\xi^3}{45}+O(\xi^5), \qquad \boxed{L(\xi)=\frac{\xi}{3}-\frac{\xi^3}{45}+O(\xi^5)} \tag{10.3.8}$$

The leading behaviour $L\approx\xi/3$ is the statement that in weak fields the alignment is proportional to the field — linear response — and the factor $1/3$ is the average of $\cos^2\theta$ over a sphere, i.e. the fact that only one of three equivalent directions is favoured.

(b) Strong fields. Write $\coth\xi = \dfrac{e^{\xi}+e^{-\xi}}{e^{\xi}-e^{-\xi}} = \dfrac{1+e^{-2\xi}}{1-e^{-2\xi}}$. For $\xi\gg 1$ the exponentials are negligible and $\coth\xi\to 1$, so

$$L(\xi)\to 1-\frac{1}{\xi} \qquad (\xi\gg 1). \tag{10.3.9}$$

Saturation: all moments aligned, $\langle\cos\theta\rangle\to 1$, approached slowly (as $1/\xi$) because thermal agitation keeps knocking moments off the axis.

(c) The identity that produces the form of Eq. (SM-2). Since $\coth\xi = 1/\tanh\xi$,

$$L(\xi) = \frac{1}{\tanh\xi}-\frac{1}{\xi} = \frac{\xi-\tanh\xi}{\xi\tanh\xi}, \qquad \text{equivalently} \qquad \xi L(\xi) = \frac{\xi-\tanh\xi}{\tanh\xi}. \tag{10.3.10}$$

This innocent rearrangement is precisely what will convert the result of Chapter 15 into the exact form printed as Eq. (SM-2); keep it in view.

10.4 The equilibrium magnetisation

Combining $\boldsymbol{M} = nm\langle\boldsymbol{e}\rangle$ [Eq. (4.5.1)] with (10.2.16):

$$\boxed{\boldsymbol{M}_0 = n\,m\,L(\xi)\,\hat{\boldsymbol{h}}, \qquad \hat{\boldsymbol{h}}=\boldsymbol{H}/H, \qquad \xi=\frac{mH}{kT}} \tag{10.4.1}$$

This is exactly the content of the paper's Eq. (SM-10),

$$\boldsymbol{M}_0 = nmL(\xi)\boldsymbol{H}/H, \qquad \xi = mH(t)/kT, \qquad L(\xi) = \coth\xi - \xi^{-1}, \tag{SM-10}$$

and of the first half of Bacri *et al.*'s Eq. (B-4), where the same object is written $\boldsymbol{M}_0(t) = \varphi M_s L(\xi)\hat{\boldsymbol{h}}$. The two forms agree because $nm = nM_sV = \varphi M_s$ by Eqs. (4.4.1) and the definition $\varphi = nV$.

Remark 10.4.1 (What "instantaneous equilibrium" means)*.* When the field varies in time, $\boldsymbol{M}_0(t)$ is defined by inserting the *instantaneous* $H(t)$ into (10.4.1). It is the magnetisation the suspension *would* have if it could reorganise infinitely fast, i.e. if $\tau_B = 0$. The real magnetisation $\boldsymbol{M}(t)$ lags behind it, and the whole of the negative-viscosity effect lives in that lag. Both papers stress this point in the text following (SM-10) and (B-4).

Weak-field form. Using (10.3.8) to leading order,

$$\boldsymbol{M}_0 = nm\frac{\xi}{3}\hat{\boldsymbol{h}} = \frac{nm}{3}\cdot\frac{mH}{kT}\hat{\boldsymbol{h}} = \frac{nm^2}{3kT}\boldsymbol{H} \equiv \chi\boldsymbol{H}, \qquad \chi = \frac{nm^2}{3kT}, \tag{10.4.2}$$

which is the paper's Eq. (SM-15). We have thus already derived, as a by-product, an equation that will be formally presented in its own place later. That is allowed and unavoidable: (SM-15) is nothing but the small-ξ limit of (SM-10).

Part V

Rotational Brownian Motion

Chapter 11

The Rotational Diffusion (Smoluchowski) Equation

11.1 Context and strategy

The Boltzmann distribution of Chapter 9 describes *equilibrium*. But the central question of both papers is intrinsically out of equilibrium: what happens to the orientations when the field oscillates, or when the fluid shears? We need the *dynamical* equation whose stationary solution is (9.5.1). That equation is the rotational analogue of the diffusion equation, and it is derived here from three ingredients only:

1. conservation of probability;
2. the deterministic drift produced by known torques (viscous, magnetic) in the inertialess limit established by Eq. (SM-1);
3. Fick's law for the diffusive part of the flux, with an as-yet-unknown diffusion coefficient D_r, whose value is then *fixed* by requiring consistency with Chapter 9.

Step 3 is the Einstein argument, and it is the reason τ_B ends up containing the viscosity η.

11.2 Calculus on the unit sphere

Orientation space is the surface of the unit sphere. We need the gradient and divergence *restricted to that surface*; they differ from the ordinary three-dimensional operators because there is no "radial" direction to differentiate along.

Definition 11.2.1 (Surface gradient)**.** For a function $f(\boldsymbol{e})$ defined on the unit sphere,

$$\nabla_s f \equiv \hat{\boldsymbol{\theta}}\,\frac{\partial f}{\partial \theta} + \hat{\boldsymbol{\phi}}\,\frac{1}{\sin\theta}\frac{\partial f}{\partial \phi}, \tag{11.2.1}$$

where $\hat{\boldsymbol{\theta}}$ and $\hat{\boldsymbol{\phi}}$ are the unit vectors of increasing θ and ϕ.

The factor $1/\sin\theta$ is not a convention: it is forced by the geometry established in (9.4.2). Moving through the angle $\mathrm{d}\phi$ carries one a *distance* $\sin\theta\,\mathrm{d}\phi$, and a gradient is a change per unit *distance*, so the derivative with respect to ϕ must be divided by $\sin\theta$. The same reasoning applied to θ needs no factor, because the distance is $\mathrm{d}\theta$ itself.

Equivalently and more usefully, $\nabla_s f$ is the ordinary three-dimensional gradient with its radial part projected out:

$$\nabla_s f = (1 - \boldsymbol{e}\boldsymbol{e}) \cdot \frac{\partial f}{\partial \boldsymbol{e}}, \qquad \text{i.e.} \qquad (\nabla_s f)_i = \big(\delta_{ij} - e_i e_j\big) \frac{\partial f}{\partial e_j}. \tag{11.2.2}$$

The operator $\delta_{ij} - e_i e_j$ is the projector onto the plane tangent to the sphere at $\boldsymbol{e}$: acting on any vector $\boldsymbol{A}$ it gives $\boldsymbol{A} - \boldsymbol{e}(\boldsymbol{e} \cdot \boldsymbol{A})$, which is $\boldsymbol{A}$ with its component along $\boldsymbol{e}$ removed. (Check that it is a projector: $(\delta_{ij} - e_i e_j)(\delta_{jk} - e_j e_k) = \delta_{ik} - e_i e_k - e_i e_k + e_i e_k (e_j e_j) = \delta_{ik} - e_i e_k$, using $e_j e_j = 1$. ✓)

Two special cases will be used constantly:

$$\nabla_s\big(\boldsymbol{e} \cdot \boldsymbol{A}\big) = \boldsymbol{A} - \boldsymbol{e}\,(\boldsymbol{e} \cdot \boldsymbol{A}) \qquad \text{for a constant vector } \boldsymbol{A}, \tag{11.2.3}$$

$$\nabla_s e_i = \hat{\boldsymbol{x}}_i - \boldsymbol{e}\, e_i. \tag{11.2.4}$$

Both follow from (11.2.2) with $\partial(\boldsymbol{e} \cdot \boldsymbol{A})/\partial e_j = A_j$ and $\partial e_i / \partial e_j = \delta_{ij}$.

Definition 11.2.2 (Surface divergence)**.** For a tangential vector field $\boldsymbol{j}(\boldsymbol{e})$ on the sphere,

$$\nabla_s \cdot \boldsymbol{j} \equiv \frac{1}{\sin\theta} \frac{\partial}{\partial \theta}\big(\sin\theta\, j_\theta\big) + \frac{1}{\sin\theta} \frac{\partial j_\phi}{\partial \phi}. \tag{11.2.5}$$

Derivation of (11.2.5)**.** Repeat the box argument of Sec. 2.3, now for a patch on the unit sphere bounded by the coordinate curves θ, $\theta + \mathrm{d}\theta$, ϕ, $\phi + \mathrm{d}\phi$. Its area is $\sin\theta\, \mathrm{d}\theta\, \mathrm{d}\phi$ by (9.4.2). The outflow through the edge at $\theta + \mathrm{d}\theta$ is (flux density)$\times$(edge length) $= j_\theta(\theta + \mathrm{d}\theta) \sin(\theta + \mathrm{d}\theta) \mathrm{d}\phi$, and the inflow at θ is $j_\theta(\theta) \sin\theta\, \mathrm{d}\phi$; their difference is $\partial_\theta\big(j_\theta \sin\theta\big) \mathrm{d}\theta\, \mathrm{d}\phi$. Note that the edge length itself depends on θ, which is why $\sin\theta$ sits *inside* the derivative. For the ϕ pair, the edge length is $\mathrm{d}\theta$ (independent of ϕ), and the net outflow is $\partial_\phi j_\phi\, \mathrm{d}\phi\, \mathrm{d}\theta$. Adding and dividing by the area gives (11.2.5). ■

Definition 11.2.3 (Surface Laplacian)**.** $\nabla_s^2 f \equiv \nabla_s \cdot (\nabla_s f) = \dfrac{1}{\sin\theta} \dfrac{\partial}{\partial\theta}\Big(\sin\theta \dfrac{\partial f}{\partial\theta}\Big) + \dfrac{1}{\sin^2\theta} \dfrac{\partial^2 f}{\partial \phi^2}$.

Result 11.2.1 (Green's identity on a closed surface)**.** *For any two smooth functions f, g on the sphere,*

$$\int f\, \nabla_s^2 g\, \mathrm{d}\Omega_e = -\int \nabla_s f \cdot \nabla_s g\, \mathrm{d}\Omega_e = \int g\, \nabla_s^2 f\, \mathrm{d}\Omega_e. \tag{11.2.6}$$

Derivation. Apply the surface divergence theorem — proved exactly as in Sec. 2.3, by summing patches and cancelling shared edges — to the tangential field $f \nabla_s g$. A closed surface has *no boundary*, so there is no boundary term at all:

$$0 = \int \nabla_s \cdot \big(f \nabla_s g\big) \mathrm{d}\Omega_e = \int \Big(\nabla_s f \cdot \nabla_s g + f \nabla_s^2 g\Big) \mathrm{d}\Omega_e, \tag{11.2.7}$$

which is the first equality; the second follows by exchanging f and g in the symmetric middle expression. ■

The absence of boundary terms is a genuine simplification peculiar to orientation space, and we shall use it without comment when integrating by parts over $\mathrm{d}\Omega_e$.

11.3 Conservation of probability

Let $W(\boldsymbol{e}, t)$ be the probability density of orientations, normalised by $\int W \mathrm{d}\Omega_e = 1$ at all times. Orientations are neither created nor destroyed: a moment can only change direction by moving continuously across the sphere. The same bookkeeping that produced the continuity equation (5.1.3) in Sec. 5.1 — now applied to a patch of the sphere instead of a box in space — gives

$$\frac{\partial W}{\partial t} + \nabla_s \cdot \boldsymbol{j} = 0, \tag{11.3.1}$$

where $\boldsymbol{j}$ is the probability flux (units: probability per unit solid angle per second, flowing tangentially).

11.4 The flux: drift

A particle whose instantaneous angular velocity is $\boldsymbol{\omega}$ carries its moment according to

$$\dot{\boldsymbol{e}} = \boldsymbol{\omega} \times \boldsymbol{e}, \tag{11.4.1}$$

which is Sec. 3.4, Step 1 applied to the unit vector $\boldsymbol{e}$ rigidly attached to the grain. (Here the "rigid dipole" assumption enters: the moment is frozen into the crystal, so rotating the grain rotates the moment. Its validity is checked numerically in Sec. 17.6.)

By Eq. (SM-1) the particle has no inertia on any relevant time scale, so its angular velocity follows instantaneously from the balance of torques:

$$0 = \underbrace{\boldsymbol{T}_{\mathrm{mag}}}_{m\,\boldsymbol{e}\times\boldsymbol{H}} \; \underbrace{-\zeta_r\big(\boldsymbol{\omega} - \boldsymbol{\Omega}\big)}_{\text{viscous, Eq. (6.5.8)}} \qquad \Longrightarrow \qquad \boldsymbol{\omega} = \boldsymbol{\Omega} + \frac{m}{\zeta_r}\big(\boldsymbol{e} \times \boldsymbol{H}\big). \tag{11.4.2}$$

Substituting into (11.4.1),

$$\dot{\boldsymbol{e}} = \boldsymbol{\Omega} \times \boldsymbol{e} + \frac{m}{\zeta_r}\big(\boldsymbol{e} \times \boldsymbol{H}\big) \times \boldsymbol{e}. \tag{11.4.3}$$

The second term simplifies beautifully. By the triple product (1.7.1) with $\boldsymbol{A} = \boldsymbol{e}$, $\boldsymbol{B} = \boldsymbol{e}$, $\boldsymbol{C} = \boldsymbol{H}$ — more directly, using $(\boldsymbol{A} \times \boldsymbol{B}) \times \boldsymbol{C} = -\boldsymbol{C} \times (\boldsymbol{A} \times \boldsymbol{B})$ —

$$\begin{aligned}
\big(\boldsymbol{e} \times \boldsymbol{H}\big) \times \boldsymbol{e} &= -\boldsymbol{e} \times \big(\boldsymbol{e} \times \boldsymbol{H}\big) && && (11.4.4)\\
&= -\Big[\boldsymbol{e}\,(\boldsymbol{e} \cdot \boldsymbol{H}) - \boldsymbol{H}\,(\boldsymbol{e} \cdot \boldsymbol{e})\Big] && \text{by (1.7.1)} && (11.4.5)\\
&= \boldsymbol{H} - \boldsymbol{e}\,(\boldsymbol{e} \cdot \boldsymbol{H}) && \text{using } \boldsymbol{e}\cdot\boldsymbol{e} = 1 && (11.4.6)\\
&= \nabla_s\big(\boldsymbol{e} \cdot \boldsymbol{H}\big) && \text{by (11.2.3).} && (11.4.7)
\end{aligned}$$

Since the orientational energy is $U(\boldsymbol{e}) = -m\,\boldsymbol{e} \cdot \boldsymbol{H}$, we have $m\nabla_s(\boldsymbol{e} \cdot \boldsymbol{H}) = -\nabla_s U$, and (11.4.3) becomes

$$\dot{\boldsymbol{e}} = \boldsymbol{\Omega} \times \boldsymbol{e} - \frac{1}{\zeta_r}\nabla_s U. \tag{11.4.8}$$

This is a completely transparent statement: the orientation is carried around by the flow, and in addition slides "downhill" on the energy landscape $U(\boldsymbol{e})$ at a rate proportional to the slope and inversely proportional to the friction. The corresponding probability flux is $W\dot{\boldsymbol{e}}$.

11.5 The flux: diffusion, and the Einstein relation

The random molecular kicks are not contained in (11.4.8). Their effect is to spread the distribution out, exactly as molecular kicks spread a dye in water. We model them by Fick's law: a flux down the gradient of concentration,

$$\boldsymbol{j}_{\text{diff}} = -D_r \nabla_s W, \tag{11.5.1}$$

where D_r is the *rotational diffusion coefficient* (units s^{-1}, since angles are dimensionless). The total flux is

$$\boldsymbol{j} = W\Big(\boldsymbol{\Omega} \times \boldsymbol{e}\Big) - \frac{W}{\zeta_r}\nabla_s U - D_r \nabla_s W. \tag{11.5.2}$$

Result 11.5.1 (Einstein relation)**.**

$$\boxed{D_r = \frac{kT}{\zeta_r}} \tag{11.5.3}$$

Derivation. The value of D_r is not free: drift and diffusion are two aspects of the same molecular bombardment, and they must be so related that the *equilibrium* distribution of Chapter 9 is reproduced. In equilibrium the fluid is at rest ($\boldsymbol{\Omega} = 0$) and nothing flows anywhere, so $\boldsymbol{j} = 0$ pointwise. Setting $\boldsymbol{\Omega} = 0$ and $W = W_0 = Ce^{-U/kT}$ in (11.5.2):

$$0 = -\frac{W_0}{\zeta_r}\nabla_s U - D_r \nabla_s W_0. \tag{11.5.4}$$

Compute $\nabla_s W_0$ by the chain rule:

$$\nabla_s W_0 = C\,\nabla_s e^{-U/kT} = C\,e^{-U/kT}\left(-\frac{\nabla_s U}{kT}\right) = -\frac{W_0}{kT}\nabla_s U. \tag{11.5.5}$$

Substituting (11.5.5) into (11.5.4),

$$0 = -\frac{W_0}{\zeta_r}\nabla_s U + \frac{D_r W_0}{kT}\nabla_s U = W_0\left(\frac{D_r}{kT} - \frac{1}{\zeta_r}\right)\nabla_s U. \tag{11.5.6}$$

Since this must hold for every orientation, and neither W_0 nor $\nabla_s U$ vanishes identically, the bracket must vanish: $D_r/kT = 1/\zeta_r$, i.e. (11.5.3). ■

Inserting the Stokes value $\zeta_r = 6\eta V$ of Eq. (6.6.1),

$$D_r = \frac{kT}{6\eta V} = \frac{kT}{8\pi\eta a^3}. \tag{11.5.7}$$

Thermal energy makes orientations wander; viscosity and size resist. Note that D_r contains η in the denominator: *the same* friction coefficient governs both the systematic response to a magnetic torque and the random wandering. That is the content of the fluctuation–dissipation idea, and it is why a purely hydrodynamic quantity (η) will appear in the final magnetic relaxation time.

11.6 The Smoluchowski equation

Substituting (11.5.2) into (11.3.1) and using (11.5.3) to eliminate ζ_r in favour of D_r:

$$\boxed{\frac{\partial W}{\partial t} = -\nabla_s\cdot\Big[W\big(\boldsymbol{\Omega}\times\boldsymbol{e}\big)\Big] + D_r\,\nabla_s\cdot\left[\nabla_s W + \frac{W}{kT}\nabla_s U\right]} \tag{11.6.1}$$

This is the "Fokker–Planck equation for rotary diffusion of colloidal particles" referred to (but not written out) by Bacri *et al.* just before their Eq. (B-5). Every subsequent result about magnetisation dynamics — the phenomenological equation (SM-7), the effective-field equation (B-6), the exact weak-field results — is a consequence of (11.6.1).

Consistency check. With $\boldsymbol{\Omega} = 0$, the stationary solution is obtained by setting the bracket to zero, which by (11.5.5) is solved by $W = Ce^{-U/kT}$: the Boltzmann distribution (9.5.1) is recovered. ✓

Chapter 12

Free Relaxation of the Magnetisation and the Value of τ_B

12.1 The question

Switch off the field and the flow. An initially aligned assembly of moments randomises. How fast? The answer defines the Brownian rotational relaxation time and is the content of Eq. (SM-5); we derive its *value* here, and present the equation formally in Chapter 17, in the papers' own order.

12.2 The equation of motion for $\langle \boldsymbol{e} \rangle$

With $\boldsymbol{\Omega} = 0$ and $U = 0$ the Smoluchowski equation (11.6.1) reduces to pure diffusion on the sphere,

$$\frac{\partial W}{\partial t} = D_r \nabla_s^2 W. \tag{12.2.1}$$

Multiply by e_i and integrate over all orientations:

$$\frac{\mathrm{d}\langle e_i \rangle}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t} \int e_i W \, \mathrm{d}\Omega_e = \int e_i \frac{\partial W}{\partial t} \, \mathrm{d}\Omega_e \qquad (e_i \text{ is a fixed label, not a function of } t) \tag{12.2.2}$$

$$= D_r \int e_i \, \nabla_s^2 W \, \mathrm{d}\Omega_e \tag{12.2.3}$$

$$= D_r \int W \, \nabla_s^2 e_i \, \mathrm{d}\Omega_e \qquad \text{by Green's identity (11.2.6).} \tag{12.2.4}$$

Everything now depends on one small calculation.

Result 12.2.1.

$$\nabla_s^2 e_i = -2 e_i. \tag{12.2.5}$$

Derivation. By isotropy it suffices to prove it for the component along the polar axis, $e_z = \cos\theta$; the result then holds for any component because we may always choose the polar axis along the direction in question. With $f = \cos\theta$ (no ϕ dependence, so the second term of the surface Laplacian vanishes):

$$\frac{\partial f}{\partial \theta} = -\sin\theta, \tag{12.2.6}$$

$$\sin\theta\frac{\partial f}{\partial\theta} = -\sin^2\theta, \tag{12.2.7}$$

$$\frac{\partial}{\partial\theta}\Big(\sin\theta\frac{\partial f}{\partial\theta}\Big) = \frac{\partial}{\partial\theta}\big(-\sin^2\theta\big) = -2\sin\theta\cos\theta, \tag{12.2.8}$$

$$\nabla_s^2 f = \frac{1}{\sin\theta}\big(-2\sin\theta\cos\theta\big) = -2\cos\theta = -2f. \quad \blacksquare \tag{12.2.9}$$

(In the language of spherical harmonics, $\boldsymbol{e}$ is a set of three $\ell = 1$ harmonics and the eigenvalue of ∇_s^2 is $-\ell(\ell+1) = -2$. We have derived the special case we need without invoking that theory.)

12.3 The relaxation law and τ_B

Substituting (12.2.5) into (12.2.4),

$$\frac{\mathrm{d}\langle e_i\rangle}{\mathrm{d}t} = D_r\int W(-2e_i)\,\mathrm{d}\Omega_e = -2D_r\langle e_i\rangle, \tag{12.3.1}$$

that is, in vector form and in terms of the magnetisation $\boldsymbol{M} = nm\langle\boldsymbol{e}\rangle$,

$$\frac{\mathrm{d}\boldsymbol{M}}{\mathrm{d}t} = -2D_r\boldsymbol{M} = -\frac{\boldsymbol{M}}{\tau_B}, \qquad \boxed{\tau_B \equiv \frac{1}{2D_r}} \tag{12.3.2}$$

By Sec. 7.4 the solution is $\boldsymbol{M}(t) = \boldsymbol{M}(0)e^{-t/\tau_B}$: the magnetisation decays exponentially with time constant τ_B. Inserting the Einstein relation (11.5.3) and $\zeta_r = 6\eta V$:

$$\tau_B = \frac{1}{2D_r} = \frac{\zeta_r}{2kT} = \frac{6\eta V}{2kT}, \tag{12.3.3}$$

that is

$$\boxed{\tau_B = \frac{3\eta V}{kT}} \tag{12.3.4}$$

which is Eq. (SM-5) and the same quantity quoted in Bacri *et al.* beneath their Eq. (B-4). We shall return to it formally in Chapter 17.

Chapter 13

The Magnetisation Relaxation Equation

13.1 Context

Solving the Smoluchowski equation (11.6.1) in full generality is hard. Both papers therefore work with a *macroscopic* equation for $\boldsymbol{M}$ alone — Eq. (SM-7) in Shliomis & Morozov, Eq. (B-4) in Bacri *et al.*:

$$\frac{\mathrm{d}\boldsymbol{M}}{\mathrm{d}t} = \boldsymbol{\Theta} \times \boldsymbol{M} - \frac{1}{\tau_B}\big(\boldsymbol{M} - \boldsymbol{M}_0\big). \tag{SM-7}$$

This chapter derives it, states precisely in which regime it is exact, and identifies exactly where it becomes an approximation — the very point on which Bacri *et al.* later base their criticism and their use of the effective-field method.

13.2 Derivation from the Smoluchowski equation

Take moments of (11.6.1), i.e. multiply by e_i and integrate. Using probability conservation in the form $\partial_t W = -\nabla_s \cdot \boldsymbol{j}$ and integrating by parts (no boundary terms, Sec. 11.2):

$$\frac{\mathrm{d}\langle e_i\rangle}{\mathrm{d}t} = -\int e_i\, \nabla_s \cdot \boldsymbol{j}\, \mathrm{d}\Omega_e = \int \boldsymbol{j} \cdot \nabla_s e_i\, \mathrm{d}\Omega_e, \tag{13.2.1}$$

with $\nabla_s e_i = \hat{\boldsymbol{x}}_i - \boldsymbol{e}\, e_i$ from (11.2.4). We treat the three contributions to $\boldsymbol{j}$ in (11.5.2) one at a time.

(i) The flow term.

$$\int W(\boldsymbol{\Omega}\times\boldsymbol{e})\cdot\big(\hat{\boldsymbol{x}}_i - \boldsymbol{e}e_i\big)\mathrm{d}\Omega_e = \int W\Big[(\boldsymbol{\Omega}\times\boldsymbol{e})_i - e_i\,(\boldsymbol{\Omega}\times\boldsymbol{e})\cdot\boldsymbol{e}\Big]\mathrm{d}\Omega_e \tag{13.2.2}$$

$$= \int W(\boldsymbol{\Omega}\times\boldsymbol{e})_i\,\mathrm{d}\Omega_e \qquad \text{by (1.5.7)} \tag{13.2.3}$$

$$= \big(\boldsymbol{\Omega}\times\langle\boldsymbol{e}\rangle\big)_i. \tag{13.2.4}$$

The averaging passed through the cross product because $\boldsymbol{\Omega}$ is a constant on the scale of the orientational average.

(ii) The diffusion term.

$$\int (-D_r \nabla_s W)\cdot \nabla_s e_i \,\mathrm{d}\Omega_e = +D_r \int W \nabla_s^2 e_i \,\mathrm{d}\Omega_e \qquad \text{Green's identity (11.2.6)} \tag{13.2.5}$$

$$= -2D_r \langle e_i \rangle \qquad \text{by (12.2.5).} \tag{13.2.6}$$

(iii) The magnetic term. Using $-\nabla_s U/\zeta_r = (m/\zeta_r)\nabla_s(\boldsymbol{e}\cdot\boldsymbol{H})$ and (11.2.3):

$$\int W \frac{m}{\zeta_r}\Big[\boldsymbol{H} - \boldsymbol{e}(\boldsymbol{e}\cdot\boldsymbol{H})\Big]\cdot\big(\hat{\boldsymbol{x}}_i - \boldsymbol{e}e_i\big)\mathrm{d}\Omega_e = \frac{m}{\zeta_r}\int W\Big[H_i - e_i(\boldsymbol{e}\cdot\boldsymbol{H})\Big]\mathrm{d}\Omega_e \tag{13.2.7}$$

$$= \frac{m}{\zeta_r}\Big[H_i - \big\langle e_i\, e_j\big\rangle H_j\Big]. \tag{13.2.8}$$

(The projector acts twice and, being idempotent, gives the same bracket; the cross-terms $\boldsymbol{e}\cdot\hat{\boldsymbol{x}}_i\, e_i(\boldsymbol{e}\cdot\boldsymbol{H})$ recombine to reproduce (13.2.7), as follows from $\boldsymbol{e}\cdot\boldsymbol{e}=1$.)

Here is the crucial structural point: the equation for the first moment $\langle e_i\rangle$ contains the *second* moment $\langle e_i e_j\rangle$. The hierarchy does not close. To obtain a closed macroscopic equation one must say something about $\langle e_i e_j\rangle$, and every such statement is a *closure approximation*.

The weak-field closure. If $\xi \ll 1$ the orientations are nearly isotropic, and for an isotropic distribution

$$\langle e_i e_j\rangle = \tfrac{1}{3}\delta_{ij} \tag{13.2.9}$$

(proved exactly as in Sec. 4.3: the average must be proportional to δ_{ij} by symmetry, and taking the trace gives $\langle e_i e_i\rangle = 1 = 3\times\tfrac{1}{3}$). Then (13.2.8) becomes

$$\frac{m}{\zeta_r}\left[H_i - \frac{H_i}{3}\right] = \frac{2m}{3\zeta_r}H_i = 2D_r\cdot\frac{mH_i}{3kT} = 2D_r\,\langle e_i\rangle_{\mathrm{eq}}, \tag{13.2.10}$$

where the Einstein relation (11.5.3) was used in the middle step and, in the last, the equilibrium weak-field average $\langle\boldsymbol{e}\rangle_{\mathrm{eq}} = L(\xi)\hat{\boldsymbol{h}} \simeq (\xi/3)\hat{\boldsymbol{h}} = m\boldsymbol{H}/3kT$ from (10.3.8).

Assembling. Adding (13.2.4), (13.2.6) and (13.2.10) in (13.2.1), and using $2D_r = 1/\tau_B$:

$$\frac{\mathrm{d}\langle\boldsymbol{e}\rangle}{\mathrm{d}t} = \boldsymbol{\Omega}\times\langle\boldsymbol{e}\rangle - \frac{1}{\tau_B}\Big(\langle\boldsymbol{e}\rangle - \langle\boldsymbol{e}\rangle_{\mathrm{eq}}\Big). \tag{13.2.11}$$

Multiplying by nm and using $\boldsymbol{M} = nm\langle\boldsymbol{e}\rangle$, $\boldsymbol{M}_0 = nm\langle\boldsymbol{e}\rangle_{\mathrm{eq}}$:

$$\boxed{\frac{\mathrm{d}\boldsymbol{M}}{\mathrm{d}t} = \boldsymbol{\Omega}\times\boldsymbol{M} - \frac{1}{\tau_B}\big(\boldsymbol{M} - \boldsymbol{M}_0\big)} \tag{13.2.12}$$

■

13.3 From $\boldsymbol{\Omega}$ to $\boldsymbol{\Theta}$: why the paper writes $\boldsymbol{\Theta}\times\boldsymbol{M}$

In (13.2.12) the rotation carrying the moments around is the *fluid* vorticity $\boldsymbol{\Omega}$, because in deriving it we split the particle's angular velocity (11.4.2) into a flow part and a magnetic part and then treated the magnetic part separately (it produced the relaxation term). An equivalent

bookkeeping keeps the *whole* mean particle angular velocity in the convective term. Since by definition the mean particle spin is

$$\boldsymbol{\Theta} = \boldsymbol{\Omega} + \frac{1}{6\eta\varphi}\big(\boldsymbol{M}\times\boldsymbol{H}\big) \tag{SM-11}$$

— which is (11.4.2) averaged over orientations, multiplied by n, and written with $n\zeta_r = 6\eta\varphi$ from Eq. (6.6.3) — the two forms differ by

$$\boldsymbol{\Theta}\times\boldsymbol{M} = \boldsymbol{\Omega}\times\boldsymbol{M} + \frac{1}{6\eta\varphi}\big(\boldsymbol{M}\times\boldsymbol{H}\big)\times\boldsymbol{M} = \boldsymbol{\Omega}\times\boldsymbol{M} - \frac{1}{6\eta\varphi}\,\boldsymbol{M}\times\big(\boldsymbol{M}\times\boldsymbol{H}\big). \tag{13.3.1}$$

Thus writing $\boldsymbol{\Theta}\times\boldsymbol{M}$, as the paper does in (SM-7), is the same as writing $\boldsymbol{\Omega}\times\boldsymbol{M}$ plus the extra nonlinear term displayed in the paper's Eq. (SM-13),

$$\frac{\mathrm{d}\boldsymbol{M}}{\mathrm{d}t} = \boldsymbol{\Omega}\times\boldsymbol{M} - \frac{1}{\tau_B}\big(\boldsymbol{M}-\boldsymbol{M}_0\big) - \frac{\tau_s}{I}\boldsymbol{M}\times\big(\boldsymbol{M}\times\boldsymbol{H}\big), \tag{SM-13}$$

since $\tau_s/I = 1/6\eta\varphi$ by Eq. (7.5.12) of Volume I. Both papers use both forms interchangeably; Bacri *et al.*'s Eq. (B-4) is exactly (SM-13) with $1/6\eta\varphi$ written out. We have now shown they are one equation.

13.4 Exactly where the approximation lies

Remark 13.4.1 (Accuracy of (SM-7)). The derivation above is *exact* to first order in ξ: the only step that required an assumption was the closure (13.2.9). For finite ξ the true second moment is not $\frac{1}{3}\delta_{ij}$ but is elongated along the field, and the relaxation term is no longer a single exponential with the single time τ_B. Consequently:

- In a *stationary* field of arbitrary strength, (SM-7) nevertheless reproduces the correct answer for the rotational viscosity remarkably well, which is why Bacri *et al.* write that "the phenomenological magnetization equation (4) is working well for arbitrary intensity of a stationary magnetic field".
- In an *alternating* field of finite amplitude and frequency, it does not. This is the failure that leads Bacri *et al.* to replace it with the effective-field equation (B-5)–(B-6) of Martsenyuk, Raikher and Shliomis, derived from the Fokker–Planck equation (11.6.1). That derivation is the business of a later volume; the point to register now is that the discrepancy between the theoretical Fig. 4 of Shliomis & Morozov and the experimental Fig. 2 of Bacri *et al.* traces back to this one closure step.

Part VI

Rotational Viscosity in a Stationary Field

Chapter 14

From Magnetic Torque to Viscosity

14.1 Context

We can now compute the magnetisation in any given flow. What we actually want is a *viscosity*. The bridge is the subject of this chapter, and it is the third of the three ingredients promised in Sec. 7.9. The logic has three links:

1. A magnetic torque acting on the suspended particles is transmitted to the liquid as a *body couple*.
2. A body couple is equivalent, in the momentum equation, to an antisymmetric stress, i.e. to the extra force density $\frac{1}{2}\nabla \times (\boldsymbol{M} \times \boldsymbol{H})$ that appears in (SM-12) and (B-3).
3. When that extra force is proportional to $\nabla^2 \boldsymbol{v}$, it simply renormalises the viscosity — and the coefficient is $\Delta\eta$.

14.2 Link 1: torque balance on the particles

The particles are inertialess (Eq. SM-1). Per unit volume of suspension, the magnetic torque density is $\boldsymbol{M} \times \boldsymbol{H}$ — this follows by summing $\boldsymbol{T} = \boldsymbol{m} \times \boldsymbol{H}$ of Eq. (4.4.2) over the n particles in unit volume and using $\boldsymbol{M} = nm\langle \boldsymbol{e} \rangle$. The viscous torque density is $-n\zeta_r(\boldsymbol{\Theta} - \boldsymbol{\Omega}) = -6\eta\varphi(\boldsymbol{\Theta} - \boldsymbol{\Omega})$ by Eq. (6.6.3). Their sum vanishes:

$$6\eta\varphi\big(\boldsymbol{\omega}_p - \boldsymbol{\Omega}\big) = \boldsymbol{M} \times \boldsymbol{H}, \tag{B-1}$$

which is Eq. (B-1) of Bacri *et al.* exactly ($\boldsymbol{\omega}_p$ is their notation for $\boldsymbol{\Theta}$). Equivalently

$$\overline{\boldsymbol{M} \times \boldsymbol{H}} = -6\eta\varphi\big(\boldsymbol{\Omega} - \boldsymbol{\Theta}\big), \tag{14.2.1}$$

the relation quoted by Shliomis & Morozov below their Eq. (SM-39). The overbar will matter only later, when the field oscillates and the torque must be averaged over the fast time; for a stationary field it may be ignored.

14.3 Link 2: a body couple is an antisymmetric stress

Result 14.3.1. *A body-couple density* $\boldsymbol{\Lambda}$ *(torque per unit volume, delivered as couples rather than as forces) is equivalent, in the linear-momentum equation, to the extra force density*

$$\boldsymbol{f}_{extra} = \tfrac{1}{2}\nabla \times \boldsymbol{\Lambda}. \tag{14.3.1}$$

Derivation. Write the stress as symmetric plus antisymmetric parts, $\sigma_{ij} = \sigma^s_{ij} + \sigma^a_{ij}$. Consider the angular-momentum balance of a small volume. The torque exerted on it by surface stresses is $\oint \varepsilon_{ijk} x_j \sigma_{kl} n_l \,\mathrm{d}S$; converting by the divergence theorem (2.3.4),

$$\oint \varepsilon_{ijk} x_j \sigma_{kl} n_l \,\mathrm{d}S = \int \partial_l\big(\varepsilon_{ijk} x_j \sigma_{kl}\big)\mathrm{d}V \tag{14.3.2}$$

$$= \int \varepsilon_{ijk}\Big(\delta_{jl}\sigma_{kl} + x_j \partial_l \sigma_{kl}\Big)\mathrm{d}V \tag{14.3.3}$$

$$= \int \varepsilon_{ijk}\sigma_{kj}\,\mathrm{d}V + \int \varepsilon_{ijk} x_j \big(\partial_l \sigma_{kl}\big)\mathrm{d}V. \tag{14.3.4}$$

The *second* integral is just $\int \boldsymbol{r} \times (\text{force density})\,\mathrm{d}V$, i.e. the torque already accounted for by the linear-momentum equation. The *first* is a genuine internal torque density, $\varepsilon_{ijk}\sigma_{kj}$, and it is nonzero only if σ has an antisymmetric part. Since the intrinsic spin angular momentum of the particles is utterly negligible (it is of order $I\Theta$, and I is the quantity shown in Volume I to be irrelevant), angular momentum balance requires this internal torque density to cancel the imposed couple:

$$\varepsilon_{ijk}\sigma^a_{kj} + \Lambda_i = 0. \tag{14.3.5}$$

Solve for σ^a with the ansatz $\sigma^a_{kj} = c\,\varepsilon_{kjl}\Lambda_l$ (the most general antisymmetric tensor built from $\boldsymbol{\Lambda}$). Then

$$\varepsilon_{ijk}\sigma^a_{kj} = c\,\varepsilon_{ijk}\varepsilon_{kjl}\Lambda_l = c\,\varepsilon_{kij}\varepsilon_{kjl}\Lambda_l \qquad \text{(cyclic: } \varepsilon_{ijk} = \varepsilon_{kij}\text{)} \tag{14.3.6}$$

$$= c\big(\delta_{ij}\delta_{jl} - \delta_{il}\delta_{jj}\big)\Lambda_l \qquad \text{by (1.6.1)} \tag{14.3.7}$$

$$= c\big(\delta_{il} - 3\delta_{il}\big)\Lambda_l = -2c\,\Lambda_i. \tag{14.3.8}$$

Comparing with (14.3.5), $-2c = -1$, so $c = \frac{1}{2}$ and

$$\sigma^a_{ij} = \tfrac{1}{2}\varepsilon_{ijl}\Lambda_l. \tag{14.3.9}$$

The force density it produces is its divergence:

$$\partial_j \sigma^a_{ij} = \tfrac{1}{2}\varepsilon_{ijl}\partial_j\Lambda_l = \tfrac{1}{2}\big(\nabla \times \boldsymbol{\Lambda}\big)_i. \quad \blacksquare \tag{14.3.10}$$

With $\boldsymbol{\Lambda} = \boldsymbol{M} \times \boldsymbol{H}$ this is exactly the last term of the paper's fluid-motion equation,

$$\rho\frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t} = -\nabla p + (\boldsymbol{M}\cdot\nabla)\boldsymbol{H} + \eta\Delta\boldsymbol{v} + \tfrac{1}{2}\nabla\times\big(\boldsymbol{M}\times\boldsymbol{H}\big), \tag{SM-12}$$

which is also Eq. (B-3). The term promised in remark 3 of Sec. 5.3 of Volume I is now derived. Note also that (SM-12) and (SM-6) are the same equation: substituting $\boldsymbol{\Theta}-\boldsymbol{\Omega} = (\tau_s/I)(\boldsymbol{M}\times\boldsymbol{H})$ from (SM-11) into the last term of (SM-6), $\frac{I}{2\tau_s}\nabla\times(\boldsymbol{\Theta}-\boldsymbol{\Omega})$, gives $\frac{1}{2}\nabla\times(\boldsymbol{M}\times\boldsymbol{H})$ identically.

14.4 Link 3: the extra force renormalises the viscosity

Assumption 14.4.1 (Proportional slip)**.** The slip between fluid and particle spins is parallel to the vorticity and proportional to it,

$$\boldsymbol{\Omega} - \boldsymbol{\Theta} = g\,\boldsymbol{\Omega}, \tag{14.4.1}$$

with g a dimensionless scalar, constant throughout the flow. This holds whenever the field is uniform and the orientational statistics are the same everywhere — true in all situations considered in these papers.

Substituting (14.2.1) and (14.4.1) into the last term of (SM-12):

$$\tfrac{1}{2}\nabla\times\big(\boldsymbol{M}\times\boldsymbol{H}\big)=\tfrac{1}{2}\nabla\times\Big[-6\eta\varphi\, g\,\boldsymbol{\Omega}\Big] \tag{14.4.2}$$

$$=-3\eta\varphi g\,\nabla\times\boldsymbol{\Omega} \tag{14.4.3}$$

$$=-3\eta\varphi g\,\nabla\times\big(\tfrac{1}{2}\nabla\times\boldsymbol{v}\big) \qquad \text{by (3.4.1)} \tag{14.4.4}$$

$$=-\tfrac{3}{2}\eta\varphi g\,\nabla\times\big(\nabla\times\boldsymbol{v}\big) \tag{14.4.5}$$

$$=-\tfrac{3}{2}\eta\varphi g\Big[\nabla(\nabla\cdot\boldsymbol{v})-\nabla^2\boldsymbol{v}\Big] \qquad \text{by (2.6.1)} \tag{14.4.6}$$

$$=+\tfrac{3}{2}\eta\varphi g\,\nabla^2\boldsymbol{v}, \qquad \text{incompressibility (5.1.4)} \tag{14.4.7}$$

so that the two viscous-looking terms of (SM-12) combine:

$$\eta\nabla^2\boldsymbol{v}+\tfrac{3}{2}\eta\varphi g\,\nabla^2\boldsymbol{v}=\Big(\eta+\tfrac{3}{2}\eta\varphi g\Big)\nabla^2\boldsymbol{v}. \tag{14.4.8}$$

The fluid therefore behaves exactly like an ordinary Newtonian fluid whose viscosity is $\eta+\Delta\eta$ with

$$\boxed{\Delta\eta=\frac{3}{2}\,\eta\varphi\, g=\frac{3}{2}\,\eta\varphi\,\frac{\Omega-\Theta}{\Omega}} \tag{14.4.9}$$

This is simultaneously Eq. (SM-34) of Shliomis & Morozov,

$$\Delta\eta=\frac{3}{2}\eta\varphi\frac{\Omega-\Theta}{\Omega}, \tag{SM-34}$$

and Eq. (B-2) of Bacri *et al.*,

$$\Delta\eta=\frac{3}{2}\eta\varphi\frac{\Omega-\omega_p}{\Omega}. \tag{B-2}$$

The identical grouping appears as Bacri *et al.*'s Eq. (B-10),

$$\eta\Delta\boldsymbol{v}-3\eta\varphi g\,\operatorname{curl}\boldsymbol{\Omega}=-2\eta\Big(1+\tfrac{3}{2}\varphi g\Big)\operatorname{curl}\boldsymbol{\Omega}, \tag{B-10}$$

which is (14.4.7) and (14.4.8) written with $\nabla^2\boldsymbol{v}=-\nabla\times(\nabla\times\boldsymbol{v})=-2\operatorname{curl}\boldsymbol{\Omega}$.

14.5 Physical interpretation of the master formula

Equation (14.4.9) is the conceptual heart of both papers, and it deserves to be read slowly.

- **If the particles spin with the fluid** ($\Theta=\Omega$, $g=0$) there is no extra viscosity. Nothing is being resisted; the suspension flows as if the grains were tiny pieces of the liquid.
- **If the particles are held still** by a strong field ($\Theta=0$, $g=1$) then $\Delta\eta=\frac{3}{2}\eta\varphi$: the maximum possible magnetoviscous increase. The liquid must flow *past* frozen obstacles, which dissipates extra energy.
- **If the particles spin faster than the fluid** ($\Theta>\Omega$, $g<0$) then $\Delta\eta<0$. The grains are now *driving* the flow rather than resisting it. This is the negative-viscosity effect, and (14.4.9) shows that it is not a paradox at all: it is what the same formula says when the sign of the slip is reversed. The energy comes from the alternating field, not from nowhere.

Equation (SM-2), to which we now turn, is nothing but (14.4.9) evaluated for a stationary field.

Chapter 15

Equation (SM-2): Rotational Viscosity in a Stationary Field

The equation to be derived is

$$\Delta\eta = \frac{3}{2}\,\eta\varphi\,\frac{\xi - \tanh\xi}{\xi + \tanh\xi}\,\sin^2\beta. \tag{SM-2}$$

15.1 Context

Shliomis & Morozov open by recalling the known behaviour of a ferrofluid in a *stationary* field: the viscosity increases. Equation (SM-2), derived by Shliomis in 1972 [their Ref. 2], is the quantitative statement of that fact. It is the reference against which the entire alternating-field analysis is set: the paper's central claim is that when the field oscillates fast enough, the same formalism produces a $\Delta\eta$ of the *opposite* sign. Bacri *et al.* quote the same physics in their Eq. (B-2) and use the measured saturation value $\eta_r(\infty, 0) \simeq 2.5$ as the calibration of their experiment.

15.2 Symbols

Symbol	Meaning	Type	Units
$\Delta\eta$	rotational (vortical) viscosity: the field-induced addition to the viscosity	scalar	P
η	viscosity of the suspension in zero field (the reference)	scalar	P
$\varphi = nV$	volume fraction of solid particles	dimensionless	—
$\xi = mH/kT$	Langevin parameter	dimensionless	—
$\tanh\xi$	hyperbolic tangent, $(e^{\xi} - e^{-\xi})/(e^{\xi} + e^{-\xi})$	function	—
β	angle between $\boldsymbol{H}$ and $\boldsymbol{\Omega}$	scalar	rad
$\boldsymbol{\Omega}, \boldsymbol{\Theta}$	angular velocities of fluid and particles	vectors	s^{-1}
$\boldsymbol{M}_0 = nmL(\xi)\hat{\boldsymbol{h}}$	equilibrium magnetisation	vector	G
$\boldsymbol{m}_1$	first-order (in $\Omega\tau_B$) correction to $\boldsymbol{M}$	vector	G
$\zeta \equiv \tau_B H M_0/6\eta\varphi$	dimensionless combination appearing in the solution	scalar	—

15.3 Assumptions

1. The field is *stationary*: $\boldsymbol{H}$ constant in time and uniform in space.
2. Rigid dipoles: the moment is frozen in the crystal (checked in Sec. 17.6).
3. The magnetisation obeys the relaxation equation (13.2.12), with its closure limitations as discussed in Sec. 13.4.
4. The flow is slow in the sense $\Omega\tau_B \ll 1$, so that the deviation of $\boldsymbol{M}$ from $\boldsymbol{M}_0$ may be treated to *first order* only. Bacri *et al.* verify this explicitly: $\tau_B \simeq 1.6$ ms and $\Omega < 60\ \mathrm{s}^{-1}$, hence $\Omega\tau_B \lesssim 0.1$.
5. Steady state: $\mathrm{d}\boldsymbol{M}/\mathrm{d}t = 0$. The field does not change and the flow is stationary, so after a few τ_B the magnetisation settles.
6. Interparticle magnetic and hydrodynamic interactions are neglected; the effect is computed per particle and multiplied by n. (Shliomis & Morozov note that $\Delta\eta$ and the Einstein correction $\frac{5}{2}\eta\varphi$ are of the same order, i.e. that we are in the dilute-suspension regime.)

15.4 Mathematical Background

Everything required has been built: the master formula (14.4.9), the relaxation equation (13.2.12), the definition (11.4.2) of $\boldsymbol{\Theta}$, the Langevin function (10.2.16) and the identity (10.3.10). One new piece of geometry is needed.

Choice of axes. Let the vorticity define the z axis and let the field lie in the xz plane at angle β from it:

$$\boldsymbol{\Omega} = \Omega\,\hat{\boldsymbol{z}}, \qquad \hat{\boldsymbol{h}} = \big(\sin\beta,\, 0,\, \cos\beta\big). \tag{15.4.1}$$

This is fully general: any two vectors span a plane, which we call xz. Two elementary cross products will be needed:

$$\hat{\boldsymbol{z}} \times \hat{\boldsymbol{h}} = \hat{\boldsymbol{z}} \times \big(\sin\beta\,\hat{\boldsymbol{x}} + \cos\beta\,\hat{\boldsymbol{z}}\big) = \sin\beta\,(\hat{\boldsymbol{z}} \times \hat{\boldsymbol{x}}) = \sin\beta\,\hat{\boldsymbol{y}}, \tag{15.4.2}$$

$$\hat{\boldsymbol{y}} \times \hat{\boldsymbol{h}} = \sin\beta\,(\hat{\boldsymbol{y}} \times \hat{\boldsymbol{x}}) + \cos\beta\,(\hat{\boldsymbol{y}} \times \hat{\boldsymbol{z}}) = -\sin\beta\,\hat{\boldsymbol{z}} + \cos\beta\,\hat{\boldsymbol{x}}, \tag{15.4.3}$$

using $\hat{\boldsymbol{z}} \times \hat{\boldsymbol{x}} = \hat{\boldsymbol{y}}$, $\hat{\boldsymbol{y}} \times \hat{\boldsymbol{x}} = -\hat{\boldsymbol{z}}$, $\hat{\boldsymbol{y}} \times \hat{\boldsymbol{z}} = \hat{\boldsymbol{x}}$.

15.5 Step-by-Step Derivation

Step 1: write the equation to be solved. Take (13.2.12) in the $\boldsymbol{\Theta}$ form (SM-7) and set $\mathrm{d}\boldsymbol{M}/\mathrm{d}t = 0$:

$$0 = \boldsymbol{\Theta} \times \boldsymbol{M} - \frac{1}{\tau_B}\big(\boldsymbol{M} - \boldsymbol{M}_0\big), \qquad \boldsymbol{\Theta} = \boldsymbol{\Omega} + \frac{1}{6\eta\varphi}\big(\boldsymbol{M} \times \boldsymbol{H}\big). \tag{15.5.1}$$

Step 2: split $\boldsymbol{M}$ into equilibrium plus a small correction.

$$\boldsymbol{M} = M_0\hat{\boldsymbol{h}} + \boldsymbol{m}_1, \qquad M_0 = nmL(\xi), \qquad |\boldsymbol{m}_1| \sim \Omega\tau_B M_0 \ll M_0. \tag{15.5.2}$$

The ordering $|\boldsymbol{m}_1|/M_0 \sim \Omega\tau_B$ follows because the only thing driving $\boldsymbol{M}$ away from $\boldsymbol{M}_0$ is the flow, whose strength relative to the restoring relaxation is exactly $\Omega\tau_B$.

Step 3: evaluate the magnetic torque to first order.

$$\boldsymbol{M} \times \boldsymbol{H} = \big(M_0\hat{\boldsymbol{h}} + \boldsymbol{m}_1\big) \times H\hat{\boldsymbol{h}} \tag{15.5.3}$$

$$= M_0 H\,\big(\hat{\boldsymbol{h}} \times \hat{\boldsymbol{h}}\big) + H\,\big(\boldsymbol{m}_1 \times \hat{\boldsymbol{h}}\big) \tag{15.5.4}$$

$$= H\,\boldsymbol{m}_1 \times \hat{\boldsymbol{h}}. \tag{15.5.5}$$

The equilibrium part contributes *nothing*: a magnetisation parallel to the field feels no torque. This single fact is why the whole effect is proportional to the small quantity $\boldsymbol{m}_1$ — and, later, why an oscillating field can reverse its sign.

Step 4: expand $\boldsymbol{\Theta} \times \boldsymbol{M}$ and drop second-order terms. Insert (15.5.5) into $\boldsymbol{\Theta}$ and then form $\boldsymbol{\Theta} \times \boldsymbol{M}$:

$$\boldsymbol{\Theta} \times \boldsymbol{M} = \left[\Omega\hat{\boldsymbol{z}} + \frac{H}{6\eta\varphi}\big(\boldsymbol{m}_1 \times \hat{\boldsymbol{h}}\big)\right] \times \Big[M_0\hat{\boldsymbol{h}} + \boldsymbol{m}_1\Big] \tag{15.5.6}$$

$$= \Omega M_0\big(\hat{\boldsymbol{z}} \times \hat{\boldsymbol{h}}\big) + \Omega\big(\hat{\boldsymbol{z}} \times \boldsymbol{m}_1\big) + \frac{HM_0}{6\eta\varphi}\big(\boldsymbol{m}_1 \times \hat{\boldsymbol{h}}\big) \times \hat{\boldsymbol{h}} + O(m_1^2). \tag{15.5.7}$$

Now compare the orders. The retained terms are $O(\Omega M_0)$; the term $\Omega(\hat{\boldsymbol{z}} \times \boldsymbol{m}_1)$ is $O(\Omega m_1) = O(\Omega^2\tau_B M_0)$, i.e. smaller by the factor $\Omega\tau_B \ll 1$ (Assumption 4). It is therefore dropped, along with $O(m_1^2)$. This is the *only* approximation made in the algebra, and it is precisely the linearisation in the vorticity.

Step 5: simplify the third term. Using the triple product (1.7.1):

$$\big(\boldsymbol{m}_1 \times \hat{\boldsymbol{h}}\big) \times \hat{\boldsymbol{h}} = -\hat{\boldsymbol{h}} \times \big(\boldsymbol{m}_1 \times \hat{\boldsymbol{h}}\big) \tag{15.5.8}$$

$$= -\Big[\boldsymbol{m}_1\big(\hat{\boldsymbol{h}} \cdot \hat{\boldsymbol{h}}\big) - \hat{\boldsymbol{h}}\big(\hat{\boldsymbol{h}} \cdot \boldsymbol{m}_1\big)\Big] \tag{15.5.9}$$

$$= -\boldsymbol{m}_1 + \hat{\boldsymbol{h}}\big(\hat{\boldsymbol{h}} \cdot \boldsymbol{m}_1\big) \tag{15.5.10}$$

$$= -\boldsymbol{m}_1^{\perp}, \tag{15.5.11}$$

where $\boldsymbol{m}_1^{\perp}$ denotes the part of $\boldsymbol{m}_1$ perpendicular to the field. The magnetic torque acts only on the perpendicular part — again the statement that a moment along the field feels nothing.

Step 6: assemble the linearised equation. Substituting (15.5.7), (15.5.11) and (15.4.2) into (15.5.1), and noting that $\boldsymbol{M} - \boldsymbol{M}_0 = \boldsymbol{m}_1$:

$$0 = \Omega M_0 \sin\beta\,\hat{\boldsymbol{y}} - \frac{HM_0}{6\eta\varphi}\boldsymbol{m}_1^{\perp} - \frac{\boldsymbol{m}_1}{\tau_B}. \tag{15.5.12}$$

Step 7: separate parallel and perpendicular components. Decompose $\boldsymbol{m}_1 = m_1^{\parallel}\hat{\boldsymbol{h}} + \boldsymbol{m}_1^{\perp}$. The source term $\Omega M_0 \sin\beta\,\hat{\boldsymbol{y}}$ is purely perpendicular to $\hat{\boldsymbol{h}}$, because $\hat{\boldsymbol{h}}$ has no y component by our choice of axes (15.4.1): $\hat{\boldsymbol{y}} \cdot \hat{\boldsymbol{h}} = 0$. Projecting (15.5.12) onto $\hat{\boldsymbol{h}}$:

$$0 = 0 - 0 - \frac{m_1^{\parallel}}{\tau_B} \qquad \Longrightarrow \qquad m_1^{\parallel} = 0. \tag{15.5.13}$$

(The field cannot change the *magnitude* of the magnetisation to this order; it can only tilt it. Physically obvious, and here proved.)

Step 8: solve for the perpendicular part. What remains of (15.5.12) is a scalar equation along $\hat{\boldsymbol{y}}$. Write $\boldsymbol{m}_1 = m_1\hat{\boldsymbol{y}}$:

$$0 = \Omega M_0 \sin\beta - \frac{HM_0}{6\eta\varphi} m_1 - \frac{m_1}{\tau_B} \tag{15.5.14}$$

$$\Omega M_0 \sin\beta = m_1 \left[\frac{1}{\tau_B} + \frac{HM_0}{6\eta\varphi}\right] \tag{15.5.15}$$

$$= \frac{m_1}{\tau_B}\left[1 + \frac{\tau_B H M_0}{6\eta\varphi}\right]. \tag{15.5.16}$$

Define the dimensionless combination

$$\zeta \equiv \frac{\tau_B H M_0}{6\eta\varphi}, \tag{15.5.17}$$

so that

$$\boxed{m_1 = \frac{\tau_B \Omega M_0 \sin\beta}{1+\zeta}} \tag{15.5.18}$$

Step 9: evaluate ζ. This is the step where all the physics of Chapters 9–12 enters at once. Insert $\tau_B = 3\eta V/kT$ from (12.3.4), $M_0 = nmL(\xi)$ from (10.4.1), and $\varphi = nV$:

$$\zeta = \frac{1}{6\eta\varphi}\cdot\frac{3\eta V}{kT}\cdot H \cdot nmL(\xi) \tag{15.5.19}$$

$$= \frac{3\eta V\, n\, m\, L(\xi) H}{6\eta\,(nV)\,kT} \tag{15.5.20}$$

$$= \frac{3\,m\,L(\xi)\,H}{6\,kT} \qquad (\eta,\ n,\ V \text{ all cancel}) \tag{15.5.21}$$

$$= \frac{1}{2}\cdot\frac{mH}{kT}\,L(\xi) \tag{15.5.22}$$

$$= \frac{\xi L(\xi)}{2}. \tag{15.5.23}$$

The cancellation of η, n and V is remarkable and important: ζ depends on the field *only* through ξ, not on the particle size, concentration, or viscosity separately.

Step 10: compute the slip $\boldsymbol{\Omega} - \boldsymbol{\Theta}$. From (15.5.1),

$$\boldsymbol{\Omega} - \boldsymbol{\Theta} = -\frac{1}{6\eta\varphi}\big(\boldsymbol{M}\times\boldsymbol{H}\big) = -\frac{H}{6\eta\varphi}\,\boldsymbol{m}_1\times\hat{\boldsymbol{h}} = -\frac{Hm_1}{6\eta\varphi}\big(\hat{\boldsymbol{y}}\times\hat{\boldsymbol{h}}\big), \tag{15.5.24}$$

and with (15.4.3),

$$\boldsymbol{\Omega} - \boldsymbol{\Theta} = \frac{Hm_1}{6\eta\varphi}\Big[\sin\beta\,\hat{\boldsymbol{z}} - \cos\beta\,\hat{\boldsymbol{x}}\Big]. \tag{15.5.25}$$

Step 11: project onto the vorticity. The master formula (14.4.9) requires the component of the slip along $\boldsymbol{\Omega}$, i.e. along $\hat{\boldsymbol{z}}$; the $\hat{\boldsymbol{x}}$ component represents a particle spin *transverse* to the vorticity, which produces no stress opposing the shear and therefore does not renormalise the viscosity (it is a symmetric-stress effect, absent from (14.4.7) because there $\nabla\times$ acts on the component along $\boldsymbol{\Omega}$). Hence

$$g = \frac{\big(\boldsymbol{\Omega}-\boldsymbol{\Theta}\big)\cdot\hat{\boldsymbol{z}}}{\Omega} = \frac{Hm_1\sin\beta}{6\eta\varphi\,\Omega}. \tag{15.5.26}$$

Step 12: insert m_1. Using (15.5.18),

$$g = \frac{H\sin\beta}{6\eta\varphi\,\Omega}\cdot\frac{\tau_B\Omega M_0\sin\beta}{1+\zeta} \tag{15.5.27}$$

$$= \frac{\tau_B H M_0}{6\eta\varphi}\cdot\frac{\sin^2\beta}{1+\zeta} \qquad (\Omega \text{ cancels}) \tag{15.5.28}$$

$$= \frac{\zeta}{1+\zeta}\,\sin^2\beta \qquad \text{by the definition (15.5.17).} \tag{15.5.29}$$

The cancellation of Ω is the mathematical expression of the fact that $\Delta\eta$ is a genuine *viscosity*: a material constant, independent of the shear rate. It is also why the fluid remains *Newtonian*, as Bacri *et al.* verify experimentally in their Fig. 1.

Step 13: express $\zeta/(1+\zeta)$ through ξ. With $\zeta = \xi L/2$ from (15.5.23),

$$\frac{\zeta}{1+\zeta} = \frac{\xi L/2}{1+\xi L/2} \tag{15.5.30}$$

$$= \frac{\xi L/2}{\dfrac{2+\xi L}{2}} \qquad \text{(common denominator in the denominator)} \tag{15.5.31}$$

$$= \frac{\xi L(\xi)}{2+\xi L(\xi)}. \tag{15.5.32}$$

Step 14: convert to the $\tanh$ form of the paper. Use the identity (10.3.10), $\xi L(\xi) = (\xi - \tanh\xi)/\tanh\xi$. Write $T \equiv \tanh\xi$ for brevity:

$$\frac{\xi L}{2+\xi L} = \frac{\dfrac{\xi - T}{T}}{2 + \dfrac{\xi - T}{T}} \tag{15.5.33}$$

$$= \frac{\dfrac{\xi - T}{T}}{\dfrac{2T + \xi - T}{T}} \qquad \text{(common denominator } T\text{)} \tag{15.5.34}$$

$$= \frac{\xi - T}{2T + \xi - T} \qquad (T \text{ cancels}) \tag{15.5.35}$$

$$= \frac{\xi - T}{\xi + T} = \frac{\xi - \tanh\xi}{\xi + \tanh\xi}. \tag{15.5.36}$$

Step 15: assemble. Substituting (15.5.36) into (15.5.29) and then into the master formula (14.4.9):

$$g = \frac{\xi - \tanh\xi}{\xi + \tanh\xi}\,\sin^2\beta, \qquad \Delta\eta = \frac{3}{2}\eta\varphi\, g, \tag{15.5.37}$$

that is

$$\boxed{\Delta\eta = \frac{3}{2}\,\eta\varphi\,\frac{\xi - \tanh\xi}{\xi + \tanh\xi}\,\sin^2\beta} \tag{SM-2}$$

which is Eq. (SM-2) exactly, with every factor accounted for. ■

15.6 Consistency Checks

Sign. For all $\xi > 0$ one has $\tanh\xi < \xi$ (since $\tanh\xi = \xi - \xi^3/3 + \cdots$), so the numerator is positive and $\Delta\eta > 0$: a stationary field always *increases* the viscosity. Bacri *et al.* give the thermodynamic reason in one sentence: a constant field can be produced by a permanent magnet, which does no work, so it cannot pump energy into the fluid and cannot reduce the dissipation.

Field parallel to the vorticity. $\beta = 0 \Rightarrow \Delta\eta = 0$. If the moments are aligned along the rotation axis, rotating about that axis does not move them, so the field exerts no torque and there is no extra friction. Both papers state exactly this beneath their Eq. (SM-2).

Zero field. $\xi \to 0$: numerator $\to 0$, so $\Delta\eta \to 0$. ✓

Saturation. $\xi \to \infty$: $\tanh\xi \to 1$, so the fraction $\to (\xi-1)/(\xi+1) \to 1$ and $\Delta\eta \to \frac{3}{2}\eta\varphi\sin^2\beta$. ✓

Order of magnitude against experiment. For Bacri *et al.*, $\varphi = 0.20$ and $\beta = \pi/2$, so the saturation prediction is $\Delta\eta/\eta = \frac{3}{2} \times 0.20 = 0.30$. The measured saturation is $\eta_r(\infty, 0) \simeq 2.5$ — eight times larger. This is *not* a failure of the derivation but a well-understood consequence of Assumption 6: at $\varphi = 0.2$ the suspension is concentrated, and interparticle interactions amplify the magnetoviscous effect far beyond the dilute prediction. The papers handle this by treating $\eta_r(\infty, 0)$ as an experimental normalisation — which is exactly what Bacri *et al.* do when they plot $g = \eta_r(\xi, \omega\tau_B)/\eta_r(\infty, 0)$ in their Figs. 3–5. The *shape* of the curve, which is what the theory predicts reliably, is what is compared.

15.7 Physical Interpretation

Read the derivation backwards and the mechanism is transparent.

The vorticity tries to rotate every grain at Ω. Rotating the grain drags its frozen moment away from the field direction, creating the transverse magnetisation m_1 of (15.5.18). A transverse magnetisation means a magnetic torque $\boldsymbol{M} \times \boldsymbol{H}$, which acts *against* the rotation. So the grain settles at a compromise spin $\Theta < \Omega$: it turns more slowly than the liquid around it. The liquid must then shear past it, and shearing past an obstacle costs energy. That extra dissipation, expressed as an effective viscosity, is $\Delta\eta$.

The structure of the answer, $\zeta/(1+\zeta)$, is the signature of a competition between two rates: $\zeta = \tau_B/\tau_{\mathrm{mag}}$ compares the Brownian randomisation time with the time $6\eta\varphi/(HM_0)$ in which the magnetic torque would restore alignment. When $\zeta \ll 1$ (weak field) Brownian motion wins, the moments barely notice the field, and $\Delta\eta \to 0$. When $\zeta \gg 1$ (strong field) the field wins completely, the moments are locked in space, the grains cannot rotate at all ($\Theta \to 0$), and $\Delta\eta$ saturates at $\frac{3}{2}\eta\varphi$.

What if a term were removed? Delete the magnetic feedback in $\boldsymbol{\Theta}$ — i.e. set $\zeta = 0$ in the denominator of (15.5.18) while keeping it in the numerator of (15.5.29) — and one gets $\Delta\eta \propto \xi L$ growing without bound in strong fields, which is unphysical. The denominator is the statement that the grains cannot be slowed by more than 100%: Θ cannot go below zero in a stationary field.

15.8 Connection to the Next Equation

Equation (SM-2) contains two limits of practical importance, which the paper displays immediately as (SM-3) and (SM-4): the weak-field quadratic growth and the strong-field saturation. We derive them next.

Chapter 16

Equations (SM-3) and (SM-4): The Two Limits

16.1 Equation (SM-3): weak fields

$$\Delta\eta = \tfrac{1}{4}\eta\varphi\xi^2, \qquad \xi \ll 1. \tag{SM-3}$$

Context. This is the regime of small fields, and it is the form in which the alternating-field theory of Sec. II of Shliomis & Morozov is worked out analytically. Every result of that section — (SM-28), (SM-29), (SM-40) — reduces to (SM-3) at zero frequency, and the authors use that fact repeatedly as a check.

Symbols. All defined; $\beta = \pi/2$ is understood, as the authors state just before Eq. (SM-3) ("below we will assume that $\beta = \pi/2$").

Assumptions. Those of Chapter 15, plus $\xi \ll 1$, i.e. $mH \ll kT$. For the Co-ferrite grains of Bacri *et al.*, $\xi = 5.25 \times 10^{-3} H(\mathrm{Oe})$, so $\xi \ll 1$ means $H \ll 190$ Oe.

Mathematical background. The series (10.3.8) for $L(\xi)$, and the elementary expansion $\tanh\xi = \xi - \xi^3/3 + 2\xi^5/15 - \cdots$, obtained from tanh = sinh / cosh by the same division performed in Sec. 10.3.

Step-by-step derivation. Work from the L-form (15.5.32), which is algebraically simpler than the tanh form:

$$\Delta\eta = \frac{3}{2}\eta\varphi\,\frac{\xi L(\xi)}{2 + \xi L(\xi)}, \qquad \sin^2\beta = 1. \tag{16.1.1}$$

For small ξ, $L(\xi) = \xi/3 + O(\xi^3)$, hence

$$\xi L(\xi) = \frac{\xi^2}{3} + O(\xi^4), \tag{16.1.2}$$

which is small compared with 2. Therefore

$$\frac{\xi L}{2 + \xi L} = \frac{\xi^2/3}{2 + \xi^2/3} \tag{16.1.3}$$

$$= \frac{\xi^2}{3}\cdot\frac{1}{2}\cdot\frac{1}{1+\xi^2/6} \qquad \text{(factor 2 out of the denominator)} \tag{16.1.4}$$

$$= \frac{\xi^2}{6}\Big(1-\frac{\xi^2}{6}+\cdots\Big) \qquad \text{geometric expansion} \tag{16.1.5}$$

$$= \frac{\xi^2}{6}+O(\xi^4). \tag{16.1.6}$$

Hence

$$\Delta\eta = \frac{3}{2}\eta\varphi\cdot\frac{\xi^2}{6} = \frac{3\eta\varphi\xi^2}{12} = \frac{1}{4}\eta\varphi\xi^2. \tag{SM-3}$$

■

Cross-check from the tanh **form.** Using $\tanh\xi \approx \xi - \xi^3/3$:

$$\frac{\xi-\tanh\xi}{\xi+\tanh\xi} \approx \frac{\xi^3/3}{2\xi-\xi^3/3} \approx \frac{\xi^3/3}{2\xi} = \frac{\xi^2}{6}, \tag{16.1.7}$$

the same result. ✓ Note how the leading terms cancel in the numerator — this is why the effect is $O(\xi^2)$ and not $O(\xi)$.

Physical interpretation. $\Delta\eta \propto H^2$. Two factors of the field are needed: one to create the transverse magnetisation $m_1 \propto \chi H$, and one to convert it into a torque $\propto m_1 H$. A single power of H cannot appear, because reversing the field ($H \to -H$) cannot change a dissipation — $\Delta\eta$ must be an even function of $\boldsymbol{H}$. Equation (SM-3) is that symmetry statement made quantitative.

Connection. The opposite limit follows immediately.

16.2 Equation (SM-4): strong fields

$$\Delta\eta = \tfrac{3}{2}\eta\varphi, \qquad \xi \gg 1. \tag{SM-4}$$

Context. The saturation value. It is the reference against which all the paper's later results are normalised: in Sec. III the quantity plotted is $\Delta\eta(\xi,\omega\tau_B)/\Delta\eta(\infty,0)$ with $\Delta\eta(\infty,0) = \frac{3}{2}\eta\varphi$, and Bacri *et al.* do the same with their measured $\eta_r(\infty,0) = 2.5$.

Assumptions. Those of Chapter 15, plus $\xi \gg 1$ ($H \gg 190$ Oe for the experimental grains) and $\beta = \pi/2$.

Step-by-step derivation. From (10.3.9), $L(\xi) \to 1 - 1/\xi$, so $\xi L(\xi) \to \xi - 1$, which is large. Then

$$\frac{\xi L}{2+\xi L} = \frac{\xi-1}{2+\xi-1} = \frac{\xi-1}{\xi+1} \tag{16.2.1}$$

$$= \frac{1-1/\xi}{1+1/\xi} \qquad \text{(divide top and bottom by } \xi) \tag{16.2.2}$$

$$\longrightarrow 1 \qquad \text{as } \xi\to\infty. \tag{16.2.3}$$

Equivalently, from the tanh form: $\tanh\xi \to 1$, so the fraction is $(\xi-1)/(\xi+1) \to 1$ — identically the same expression, as it must be. Therefore

$$\Delta\eta \to \frac{3}{2}\eta\varphi. \tag{SM-4}$$

■

Physical interpretation. In a saturating field the moments — and, being rigid dipoles, the grains themselves — are locked in space: $\Theta = 0$. The master formula (14.4.9) with $g = 1$ then gives $\frac{3}{2}\eta\varphi$ directly, without any calculation. Equation (SM-4) is thus the statement that a strongly magnetised ferrofluid behaves like a suspension of *non-rotating* spheres.

A revealing comparison. Shliomis & Morozov point out that $\frac{3}{2}\eta\varphi$ is of the same order as the Einstein correction $\frac{5}{2}\eta\varphi$ for the ordinary (translational) viscosity of a suspension of spheres. The two are physically distinct — one comes from the particles refusing to *rotate*, the other from their refusing to *deform* — but both are first order in φ, confirming that we are consistently working to leading order in a dilute suspension, and that no larger effect has been overlooked.

Connection. Both limits are controlled by the ratio mH/kT, so the next thing the paper must specify is the *time* scale on which the moments respond, since an alternating field will be compared against it. That is Eq. (SM-5).

Chapter 17

Equation (SM-5): The Brownian Relaxation Time

$$\tau_B = 3\eta V/kT. \tag{SM-5}$$

17.1 Context

Everything so far concerned a stationary field. The paper now turns to the alternating field $\boldsymbol{H} = (H_0 \cos \omega t, 0, 0)$, and the decisive question becomes: can the magnetisation follow the field? The answer depends on comparing the field period with the time the moments need to reorganise. Equation (SM-5) is that time. The dimensionless product $\omega\tau_B$ that it defines is, together with ξ, one of the two variables in terms of which every remaining result of both papers is expressed — the axes of Figs. 3, 4 and 5 of both papers are precisely ξ and $\omega\tau_B$.

17.2 Symbols

Symbol	Meaning	Type	Units
τ_B	Brownian time of rotational diffusion: the time for the magnetisation to relax by mechanical rotation of the grains	scalar, constant	s
η	viscosity of the carrier liquid	scalar	P
$V = \pi d^3/6$	volume of one particle	scalar	cm^3
kT	thermal energy	scalar	erg
$D_r = kT/6\eta V$	rotational diffusion coefficient	scalar	s^{-1}
τ_N	Néel time (internal reorientation of the moment)	scalar	s
K	magnetic anisotropy constant of the particle material	scalar	$\mathrm{erg\,cm}^{-3}$
d_s	crossover diameter at which $\tau_N \sim \tau_B$	scalar	cm

17.3 Assumptions

1. Rigid dipole: the moment is frozen in the crystal lattice, so reorienting the moment requires physically rotating the grain (justified in Sec. 17.6).
2. The grain is a sphere obeying Stokes rotational friction, $\zeta_r = 6\eta V$.
3. The liquid is a Newtonian continuum on the scale of the grain — an idealisation, since $d = 10$ nm is only ~ 30 water molecules across, but one that experiment vindicates.
4. Dilute suspension: each grain diffuses independently.

17.4 Derivation

Fully carried out in Chapter 12. Collecting the three ingredients:

$$\text{Einstein relation (11.5.3):} \qquad D_r = \frac{kT}{\zeta_r}, \tag{17.4.1}$$

$$\text{Stokes friction (6.6.1):} \qquad \zeta_r = 6\eta V, \tag{17.4.2}$$

$$\text{Relaxation of the first moment (12.3.2):} \qquad \tau_B = \frac{1}{2D_r}, \tag{17.4.3}$$

whence

$$\tau_B = \frac{1}{2}\cdot\frac{\zeta_r}{kT} = \frac{6\eta V}{2kT} = \frac{3\eta V}{kT}. \tag{SM-5}$$

Dimensional check: $[\eta V/kT] = (\mathrm{g\,cm^{-1}s^{-1}}\cdot\mathrm{cm^3})/\mathrm{erg} = (\mathrm{g\,cm^2 s^{-1}})/(\mathrm{g\,cm^2 s^{-2}}) = \mathrm{s}$. ✓

17.5 Numerical Values

An isolated 10-nm grain in water. $d = 10^{-6}$ cm gives $V = \pi d^3/6 = \pi\times10^{-18}/6 = 5.24\times10^{-19}$ cm^3; with $\eta = 10^{-2}$ P and $kT = 1.38\times10^{-16}\times300 = 4.14\times10^{-14}$ erg,

$$\tau_B = \frac{3\times10^{-2}\times5.24\times10^{-19}}{4.14\times10^{-14}} = \frac{1.57\times10^{-20}}{4.14\times10^{-14}} = 3.8\times10^{-7}\ \mathrm{s}. \tag{17.5.1}$$

Shliomis & Morozov state that for such particles τ_B "is usually not higher than 10^{-5} s", which our value comfortably satisfies. ✓

The experimental fluid of Bacri *et al.* They *measure* $\tau_B \simeq 1.6$ ms by transient birefringence — four thousand times larger. Both factors in ηV are responsible:

- The relevant η is not that of pure water but of the concentrated suspension in which each grain finds itself; the measured zero-field value is 77 cP = 0.77 P, i.e. 77 times larger.
- $\tau_B \propto V \propto d^3$, so the *largest* grains of a polydisperse sample dominate the measured relaxation, and each grain carries a surfactant/hydration layer that increases its hydrodynamic diameter above the magnetic core diameter.

Solving $\tau_B = 3\eta V/kT$ for the effective diameter with $\eta = 0.77$ P and $\tau_B = 1.6\times10^{-3}$ s:

$$V = \frac{\tau_B kT}{3\eta} = \frac{1.6\times10^{-3}\times4.14\times10^{-14}}{2.31} = 2.9\times10^{-17}\ \mathrm{cm^3} \implies d = \left(\frac{6V}{\pi}\right)^{1/3} = 3.8\times10^{-6}\ \mathrm{cm}, \tag{17.5.2}$$

i.e. an effective hydrodynamic diameter of about 38 nm for a magnetic core of 10 nm. That is a large but entirely plausible figure for a 20% ferrofluid, in which small clusters and the surfactant shell both contribute. The important point for the theory is that τ_B enters only through the measurable combination $\omega\tau_B$, and Bacri *et al.* accordingly use their *measured* value, converting frequency to the dimensionless variable by

$$\omega\tau_B = 2\pi f \tau_B = 10^{-2} f \ \text{(Hz)}, \tag{17.5.3}$$

which is the calibration printed in their text. Check: $2\pi \times 1.6 \times 10^{-3} = 1.005 \times 10^{-2}$. ✓

Where the interesting physics lies. $\omega\tau_B = 1$ corresponds, by (17.5.3), to $f = 100$ Hz. The experiments run from 0 to 1480 Hz, i.e. from $\omega\tau_B = 0$ to $\omega\tau_B \approx 15$: exactly the range in which the sign change predicted in the following chapters occurs. This is not a coincidence — the ferrofluid was chosen so that the effect would be accessible at audio frequencies.

17.6 Physical Interpretation, and the Néel Alternative

τ_B is the time in which thermal collisions rotate a grain through an angle of order one radian. It grows with η (a stiffer liquid resists), grows with V (a bigger grain has more surface to drag), and falls with T (more vigorous kicks).

There is, however, a second way for the magnetisation to relax, which does *not* require the grain to move: the moment can rotate *inside* the crystal, jumping over the magnetic anisotropy barrier. This is the Néel mechanism, with time

$$\tau_N = \tau_0 \exp\!\left(\frac{KV}{kT}\right), \qquad \tau_0 \sim 10^{-9}\ \text{s}, \tag{17.6.1}$$

where K is the anisotropy energy per unit volume, so that KV is the height of the barrier. If $\tau_N \ll \tau_B$ the moment relaxes without the grain turning, no mechanical torque is transmitted to the liquid, and the entire magnetoviscous effect disappears. Shliomis & Morozov make this point explicitly: only particles with "frozen-in" moments contribute, and the crossover diameter d_s at which the two rates are comparable falls near the sizes commonly used in ferrofluids.

Numerical check for the experimental fluid. Bacri *et al.* chose Co-ferrite precisely to be safe on this point, quoting $K = 2 \times 10^6$ erg cm^{-3}. With $V = 5.24 \times 10^{-19}$ cm^3 for a 10 nm grain:

$$KV = 2 \times 10^6 \times 5.24 \times 10^{-19} = 1.05 \times 10^{-12}\ \text{erg}, \tag{17.6.2}$$

$$\frac{KV}{kT} = \frac{1.05 \times 10^{-12}}{4.14 \times 10^{-14}} = 25.3, \tag{17.6.3}$$

$$\tau_N = 10^{-9} \times e^{25.3} \approx 10^{-9} \times 9.8 \times 10^{10} \approx 98\ \text{s}. \tag{17.6.4}$$

The Néel time is about 100 seconds — eight orders of magnitude longer than τ_B, and eleven orders longer than the field period. The moment is, for all experimental purposes, welded to the crystal. This is exactly the claim Bacri *et al.* make when they write that the large anisotropy constant "induces a rigid dipole behaviour of the particles". Assumption 1 of this chapter, and Assumption 2 of Chapter 15, are thereby justified quantitatively.

17.7 Connection to the Next Equation

With τ_B defined, the two dimensionless variables of the entire problem are in place:

$$\xi = \frac{mH}{kT} \quad \text{(how strong the field is)}, \qquad \omega\tau_B \quad \text{(how fast it alternates)}. \tag{17.7.1}$$

The paper now writes down the complete system of ferrohydrodynamic equations that must be solved in the $(\xi, \omega\tau_B)$ plane — Eqs. (SM-6) to (SM-10), which are the fluid-motion equation, the magnetisation equation, the equation of rotational motion of the particles, the Maxwell constraints, and the Langevin formula. Every one of these five has in fact already been derived in Volumes I and II; Volume III will assemble them formally in the paper's own order, then proceed to the elimination (SM-11)–(SM-13), the alternating field (SM-14)–(SM-15), and the circular-polarisation solution beginning at (SM-16).

Chapter 18

Status and Road Map

18.1 What has been established in Volume II

Result	Statement	Where derived
Boltzmann law	$p \propto e^{-E/kT}$, from the two postulates	Sec. 9.3
Orientational equilibrium	$W_0 = \dfrac{\xi}{4\pi \sinh \xi} e^{\xi \cos\theta}$	Sec. 9.5
Langevin function	$\langle \cos\theta \rangle = L(\xi) = \coth \xi - 1/\xi$	Sec. 10.2
Series and identity	$L \simeq \xi/3 - \xi^3/45$; $L \to 1 - 1/\xi$; $\xi L = \dfrac{\xi - \tanh \xi}{\tanh \xi}$	Sec. 10.3
Eq. (SM-10)	$\boldsymbol{M}_0 = nmL(\xi)\hat{\boldsymbol{h}}$ (formally restated in Vol. III)	Sec. 10.4
Eq. (SM-15)	$\chi = nm^2/3kT$ (formally restated in Vol. III)	Sec. 10.4
Surface calculus	∇_s, $\nabla_{s'}$, ∇_s^2, Green's identity	Sec. 11.2
Smoluchowski equation	$\partial_t W = -\nabla_s \cdot [W(\boldsymbol{\Omega} \times \boldsymbol{e})] + D_r \nabla_s \cdot [\nabla_s W + W \nabla_s U/kT]$	Sec. 11.6
Einstein relation	$D_r = kT/\zeta_r = kT/6\eta V$	Sec. 11.5
Eigenvalue	$\nabla_s^2 \boldsymbol{e} = -2\boldsymbol{e}$	Sec. 12.2
Eq. (SM-7)/(B-4)	$\mathrm{d}\boldsymbol{M}/\mathrm{d}t = \boldsymbol{\Theta} \times \boldsymbol{M} - (\boldsymbol{M} - \boldsymbol{M}_0)/\tau_B$, exact for $\xi \ll 1$	Ch. 13
Eq. (SM-11)	$\boldsymbol{\Theta} = \boldsymbol{\Omega} + (\boldsymbol{M} \times \boldsymbol{H})/6\eta\varphi$	Sec. 13.3
Eq. (SM-13)	equivalent nonlinear form of (SM-7)	Sec. 13.3
Couple → stress	body couple $\boldsymbol{\Lambda} \Leftrightarrow$ force $\frac{1}{2}\nabla \times \boldsymbol{\Lambda}$	Sec. 14.3
Eq. (SM-12)/(B-3)	the ferrohydrodynamic momentum equation	Sec. 14.3
Eq. (B-1)	$6\eta\varphi(\boldsymbol{\omega}_p - \boldsymbol{\Omega}) = \boldsymbol{M} \times \boldsymbol{H}$	Sec. 14.2
Eq. (SM-34)/(B-2)/(B-10)	$\Delta\eta = \frac{3}{2}\eta\varphi(\Omega - \Theta)/\Omega$	Sec. 14.4
Eq. (SM-2)	$\Delta\eta = \frac{3}{2}\eta\varphi \dfrac{\xi - \tanh \xi}{\xi + \tanh \xi} \sin^2 \beta$	Ch. 15
Eq. (SM-3)	$\Delta\eta = \frac{1}{4}\eta\varphi\xi^2$, $\xi \ll 1$	Sec. 16.1
Eq. (SM-4)	$\Delta\eta = \frac{3}{2}\eta\varphi$, $\xi \gg 1$	Sec. 16.2
Eq. (SM-5)	$\tau_B = 3\eta V/kT$; $\omega\tau_B = 10^{-2} f(\mathrm{Hz})$ experimentally	Ch. 17

Note that several equations of the papers (SM-7, SM-10, SM-11, SM-12, SM-13, SM-15, SM-34, B-1, B-2, B-3, B-4, B-10) have been *derived* in Volume II because they were needed as tools. They will be *presented* in the eight-part format, in the papers' own order, when we reach them — with cross-references back to the derivations given here, and with no repetition of the algebra.

18.2 Next

Volume III will cover, in order:

- Eq. (SM-6): the fluid-motion equation with the spin-coupling term, in its original form;
- Eqs. (SM-7), (SM-8): the magnetisation equation and the equation of rotational motion of the particles;
- Eq. (SM-9): the Maxwell constraints $\nabla \cdot \boldsymbol{v} = 0$, $\nabla \times \boldsymbol{H} = 0$, $\nabla \cdot (\boldsymbol{H} + 4\pi \boldsymbol{M}) = 0$, each derived from the Maxwell equations for a nonconducting medium;
- Eq. (SM-10): the Langevin magnetisation, formally;
- Eqs. (SM-11)–(SM-13): elimination of the particle spin, including the explicit estimate $I\,\mathrm{d}\boldsymbol{\Theta}/\mathrm{d}t \ll I\boldsymbol{\Theta}/\tau_s$;
- Eqs. (SM-14)–(SM-15): the alternating field and the linear magnetisation law.

After that, Volume IV takes up the circular-polarisation decomposition (SM-16)–(SM-24), which is where the sign reversal of $\Delta\eta$ finally emerges.

End of Volume II. Volume III begins with Eq. (SM-6).